

MODEL BASED DESIGN TOOLBOX

ENABLING FAST PROTOTYPING AND DESIGN

- Electric Motor Classification
- Mathematical Model
- Motor operation



Motor Classification – Many Types

- AC vs DC primary difference is the type of voltage applied to the armature

Electric Motors

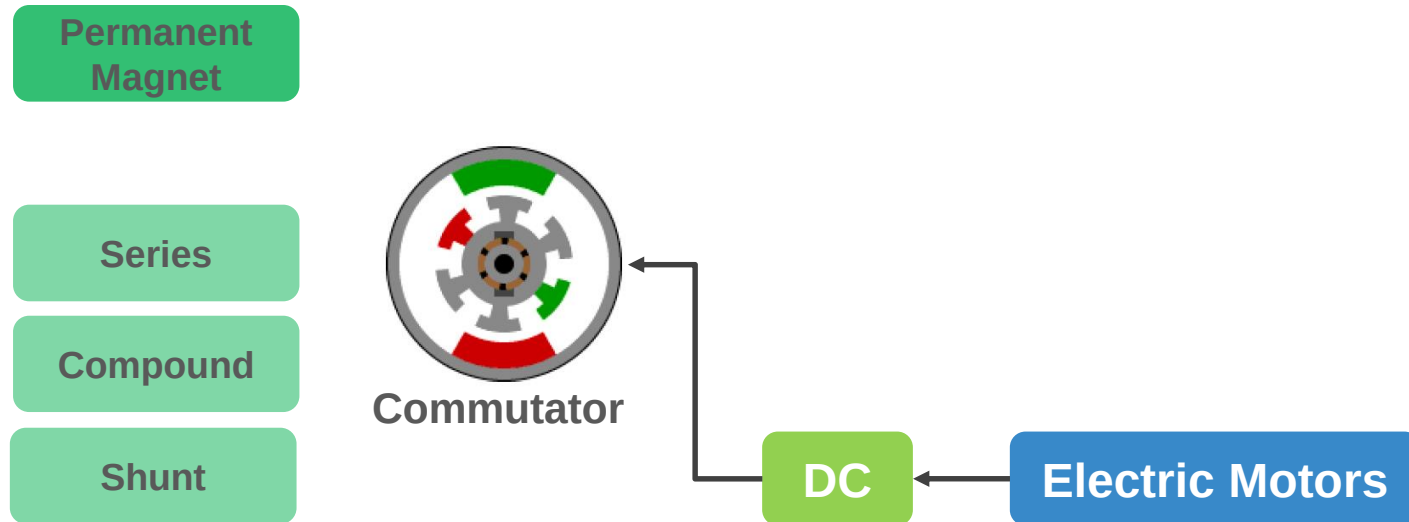
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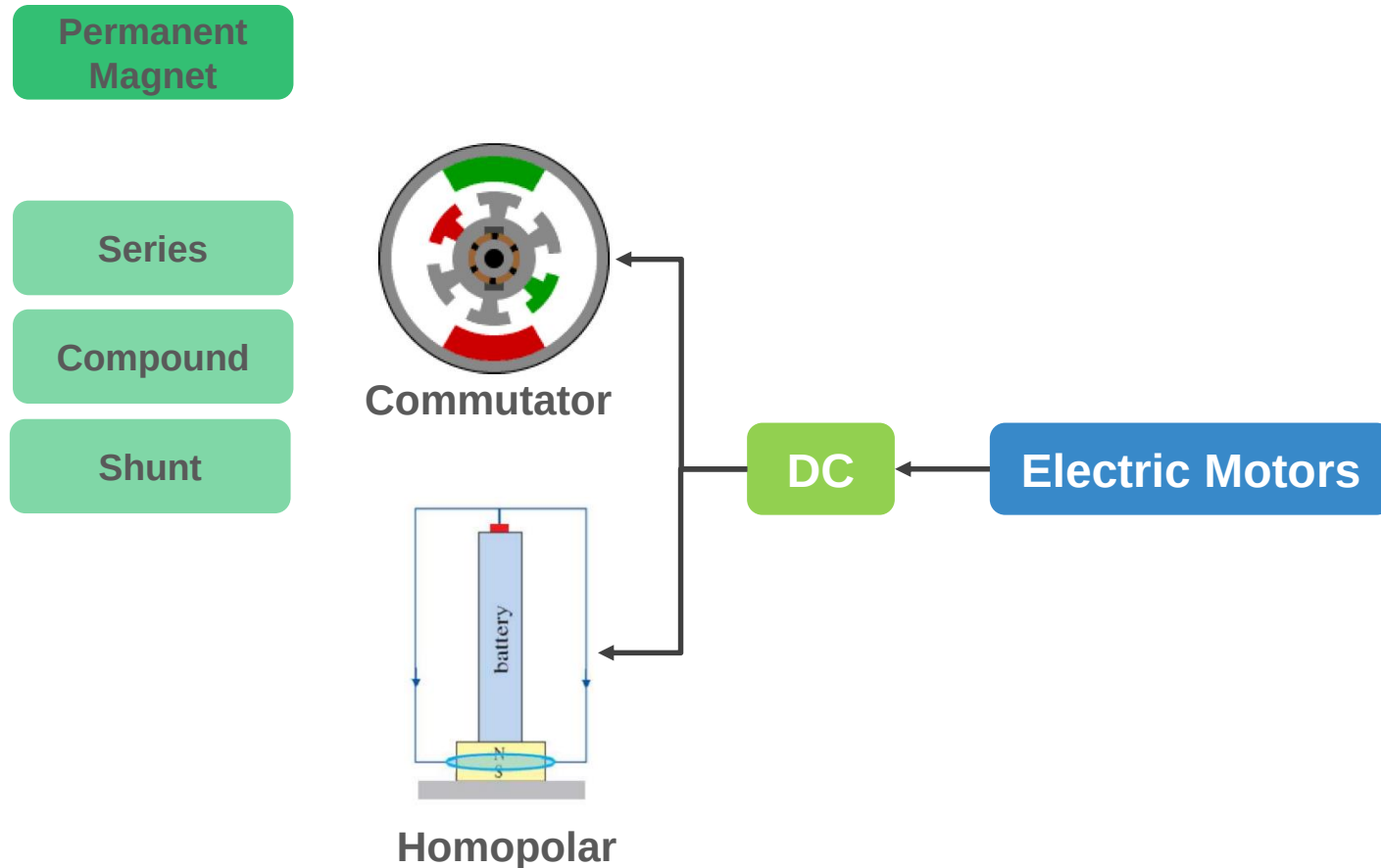
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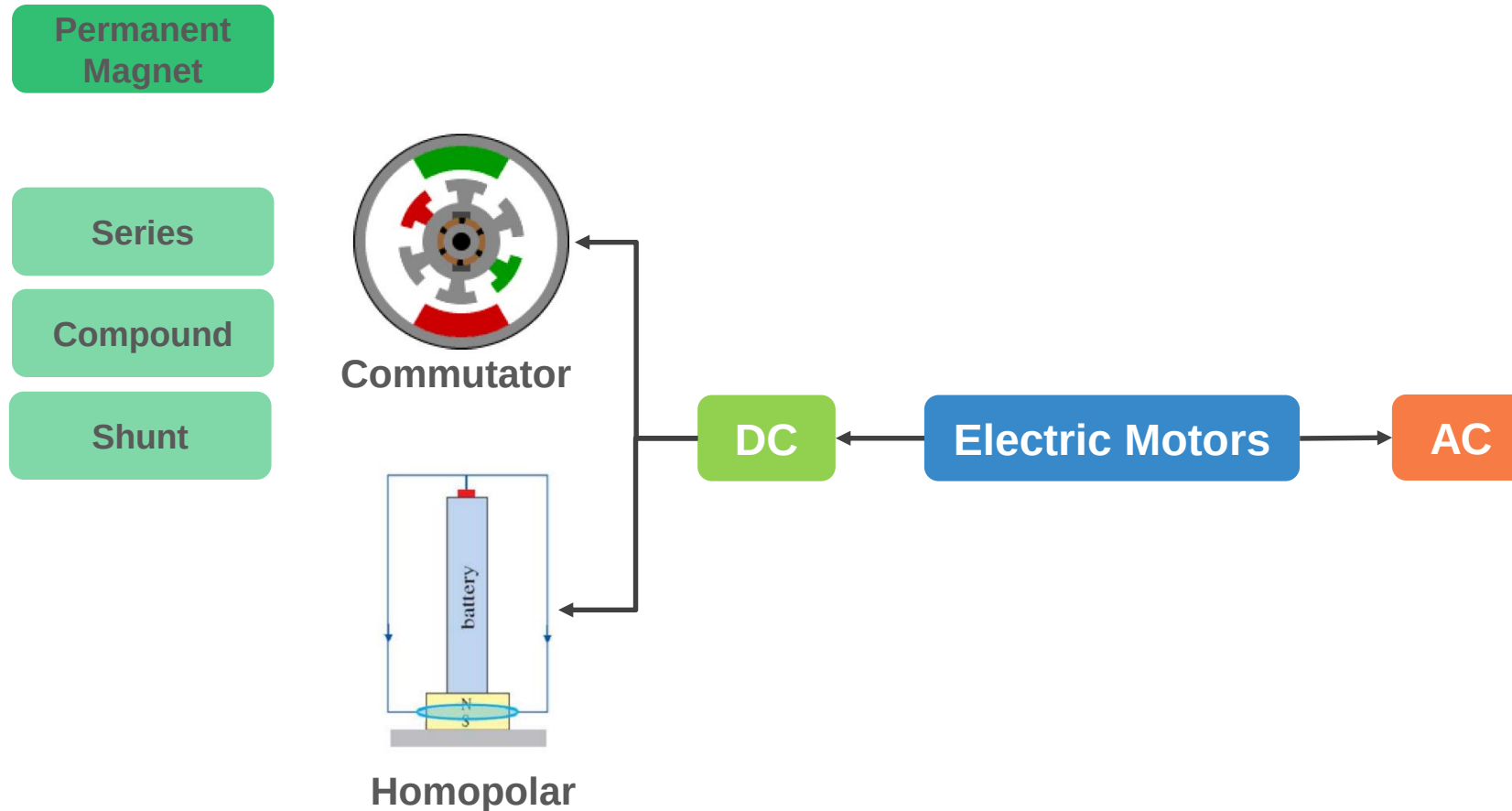
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Source: <http://vicente1064.blogspot.ro/>

Motor Classification – Many Types

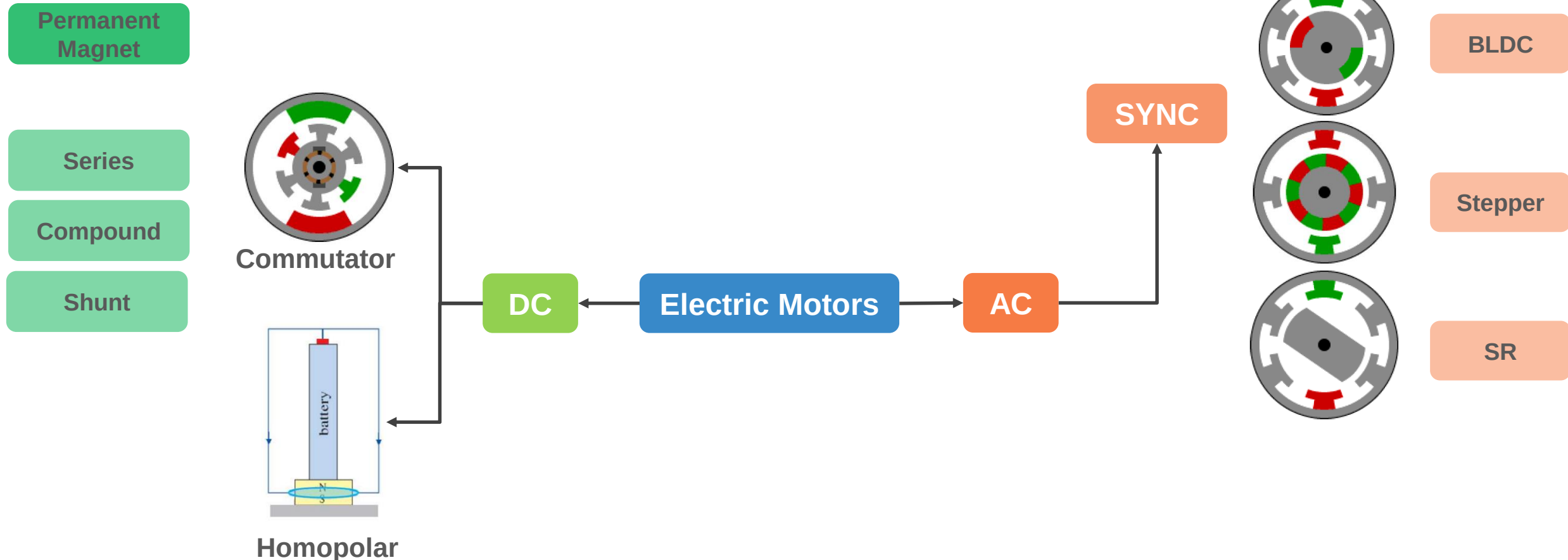
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Motor Classification – Many Types

- AC vs DC primary difference is the type of voltage applied to the armature



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Motor Classification – Many Types

- AC vs DC primary difference is the type of voltage applied to the armature
- SYNC vs ASYNC primary difference is whether or not the rotor and stator fields runs at same frequency

Permanent
Magnet

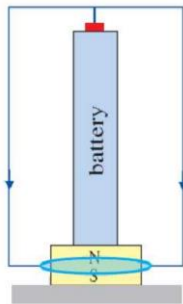
Series

Compound

Shunt



Commutator



Homopolar

Source: <http://vicente1064.blogspot.ro/>

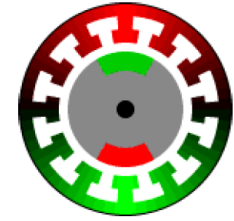
DC

Electric Motors

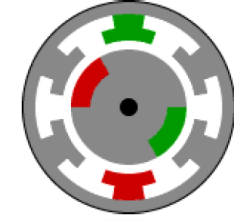
AC

SYNC

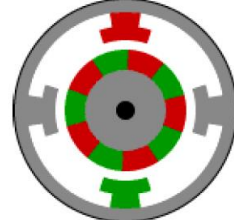
ASYNC



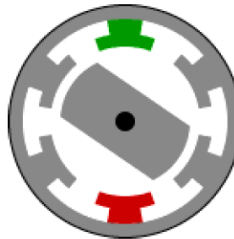
PMSM



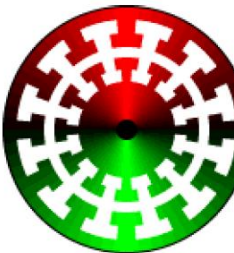
BLDC



Stepper



SR

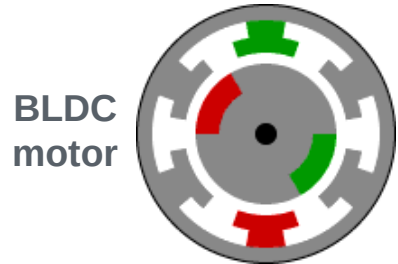
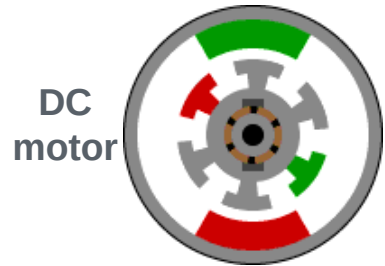


Single
phase

Poly-
phase

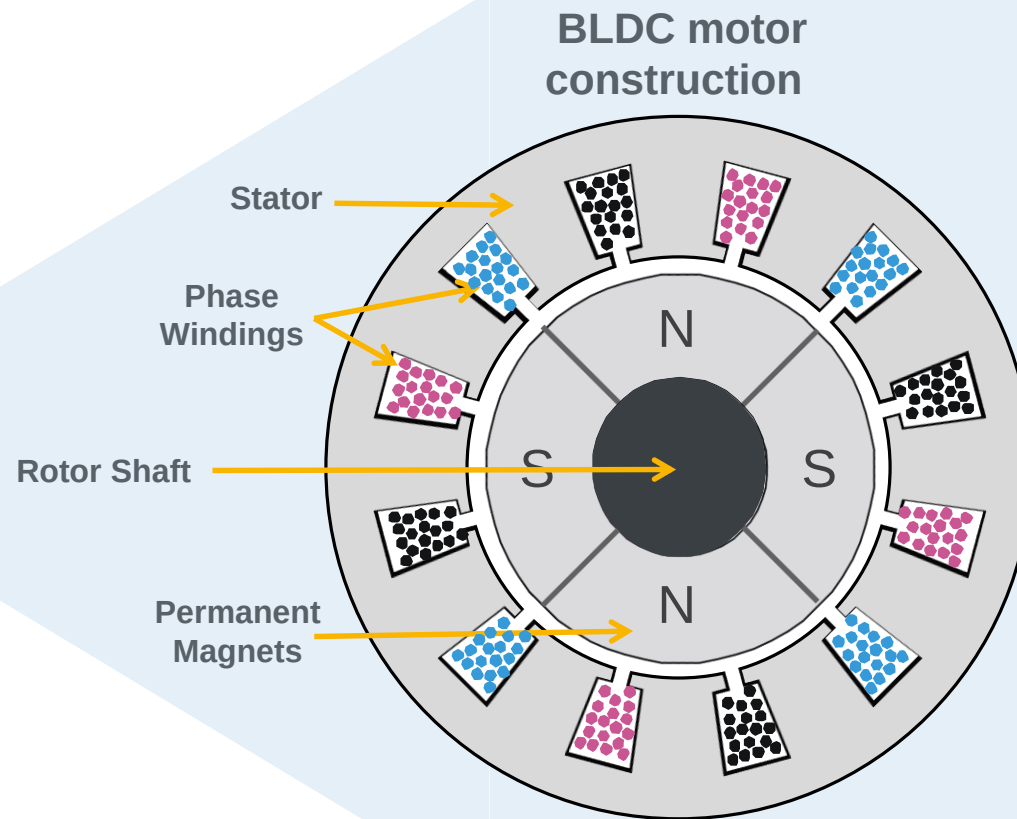
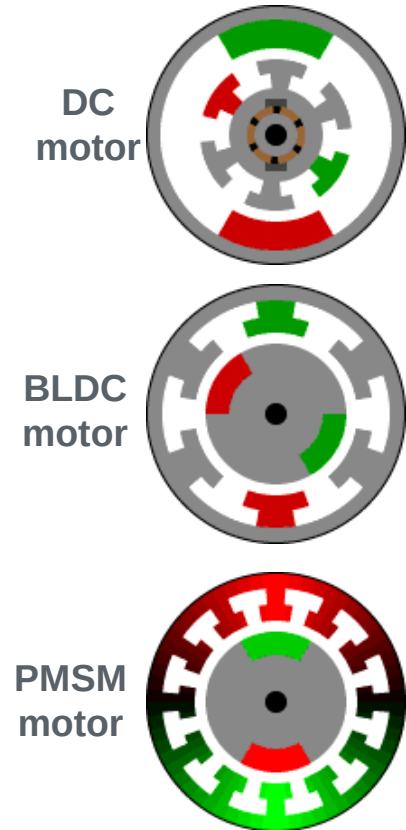
What is a BLDC motor

- Consists of a rotor with permanent magnets and a stator with phase windings.
- Needs electronic commutation for the control of current through its three phase windings.
- One phase is un-powered at any given time



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BLDC examples



BLDC Motor – Mathematical Model

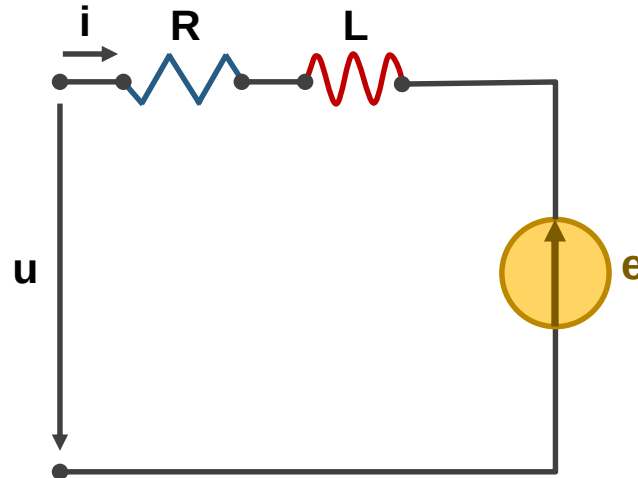
- Easy to model as 2 parts:

BLDC Motor – Mathematical Model

- Easy to model as 2 parts:
 - Electric circuit system – models the motor phase behavior

Electrical equation:

$$u = R \cdot i + L \cdot \frac{di}{dt} + e$$



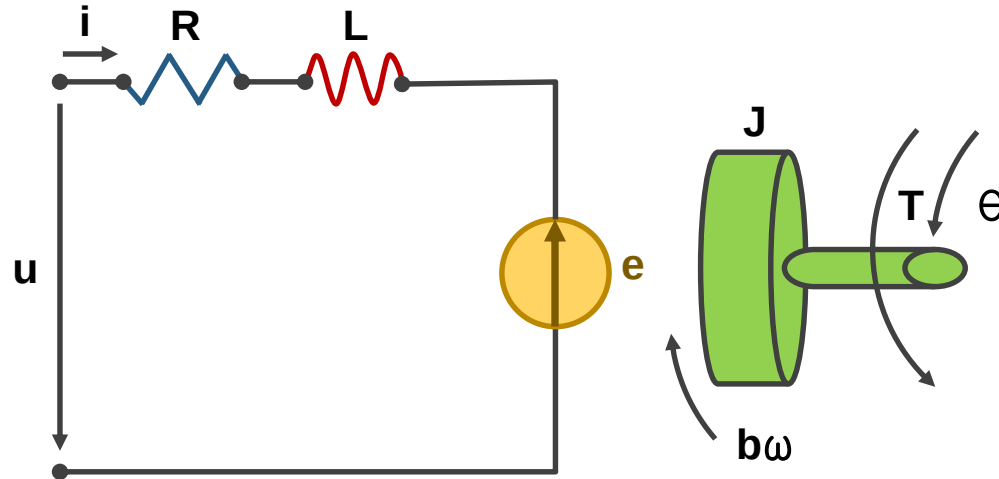
L – armature self inductance
R – armature resistance
e – back-EMF
i – phase current
u – terminal phase voltage

BLDC Motor – Mathematical Model

- Easy to model as 2 parts:
 - Electric circuit system – models the motor phase behavior
 - Mechanical system – models the rotor dynamics

Electrical equation:

$$u = R \cdot i + L \cdot \frac{di}{dt} + e$$



L – armature self inductance
R – armature resistance
e – back-EMF
i – phase current
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Mechanical equation:

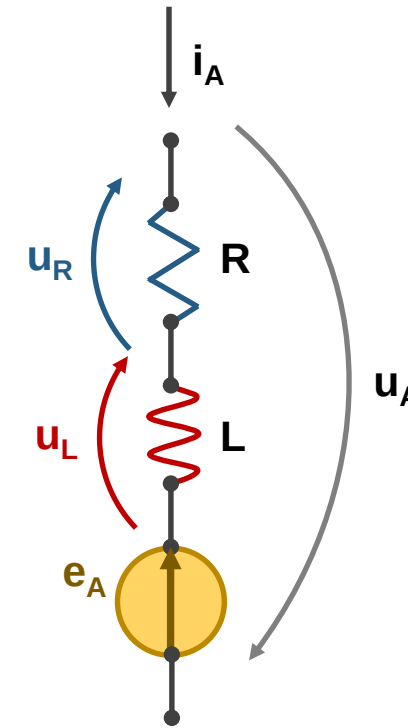
$$T = J \cdot \frac{d\omega}{dt} + b \cdot \omega$$

J – rotor inertia
b – friction constant
T – total torque
θ – rotor position
ω – rotor speed

BLDC Motor - Electrical System

- Armature winding mathematical model

$$u_A = R \cdot i_A + L \cdot \frac{di_A}{dt} + e_A$$



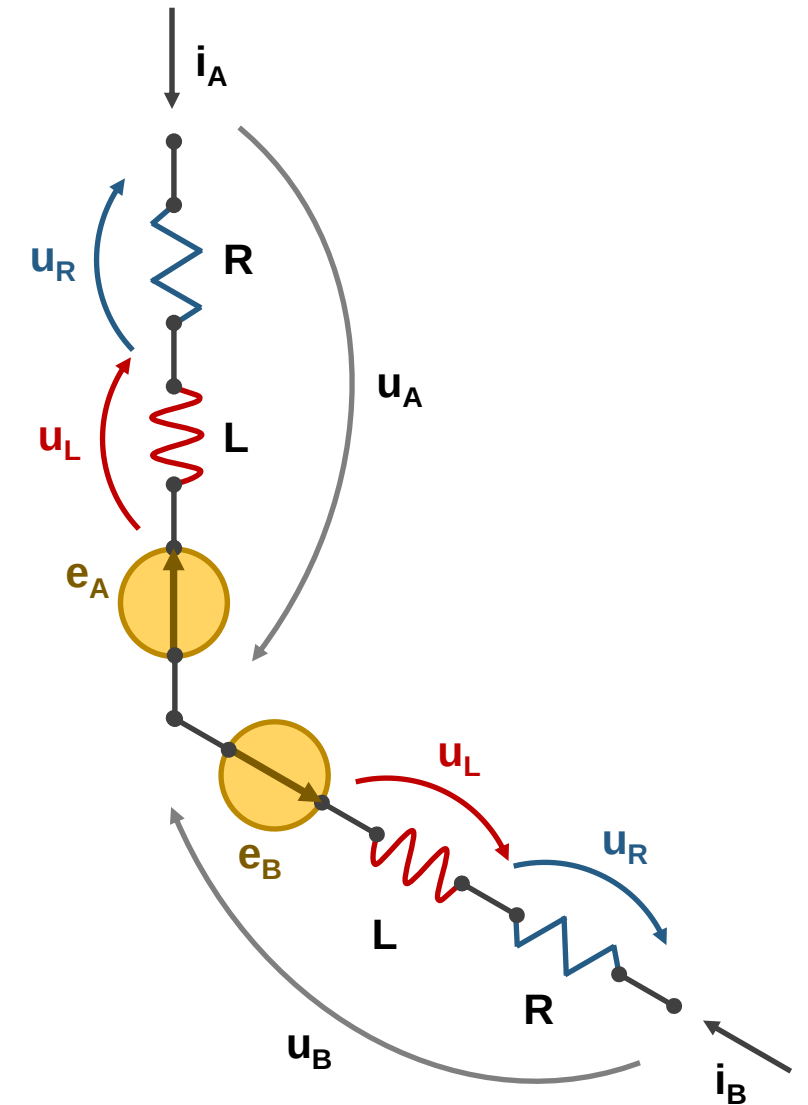
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BLDC Motor - Electrical System

- Armature winding mathematical model

$$u_A = R \cdot i_A + L \cdot \frac{di_A}{dt} + e_A$$

$$u_B = R \cdot i_B + L \cdot \frac{di_B}{dt} + e_B$$



L – armature self inductance
R – armature resistance
e – back-EMF
i – phase current
u – terminal phase voltage



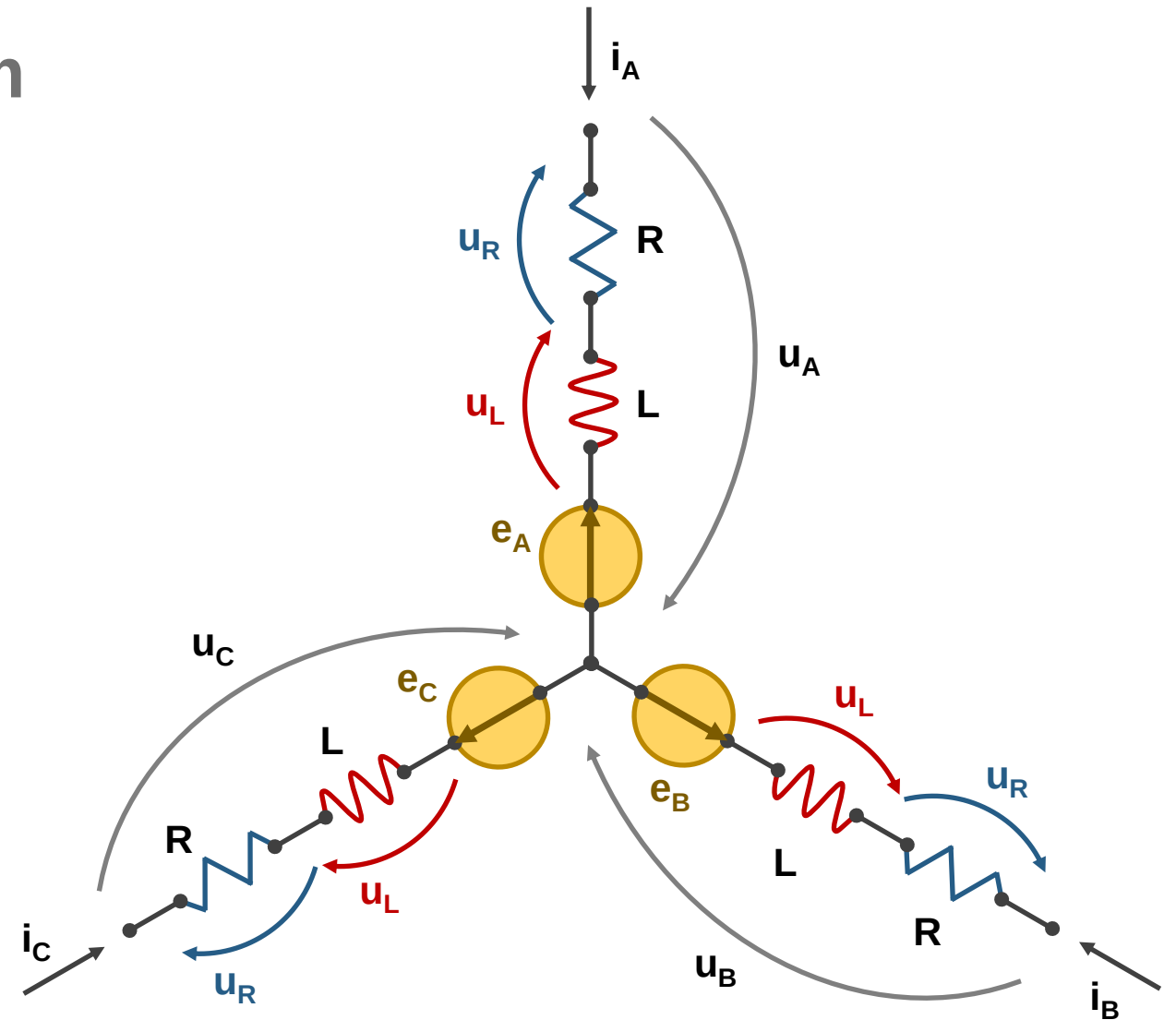
BLDC Motor - Electrical System

- Armature winding mathematical model

$$u_A = R \cdot i_A + L \cdot \frac{di_A}{dt} + e_A$$

$$u_B = R \cdot i_B + L \cdot \frac{di_B}{dt} + e_B$$

$$u_C = R \cdot i_C + L \cdot \frac{di_C}{dt} + e_C$$



L – armature self inductance
R – armature resistance
e – back-EMF
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BLDC Motor - Electrical System

- Armature winding mathematical model

$$u_A = R \cdot i_A + L \cdot \frac{di_A}{dt} + e_A$$

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$$u_C = R \cdot i_C + L \cdot \frac{di_C}{dt} + e_C$$

- Back-EMF is dependent of rotor position and armature winding

$$e_A = K_w \cdot f(\theta_e) \cdot \omega$$

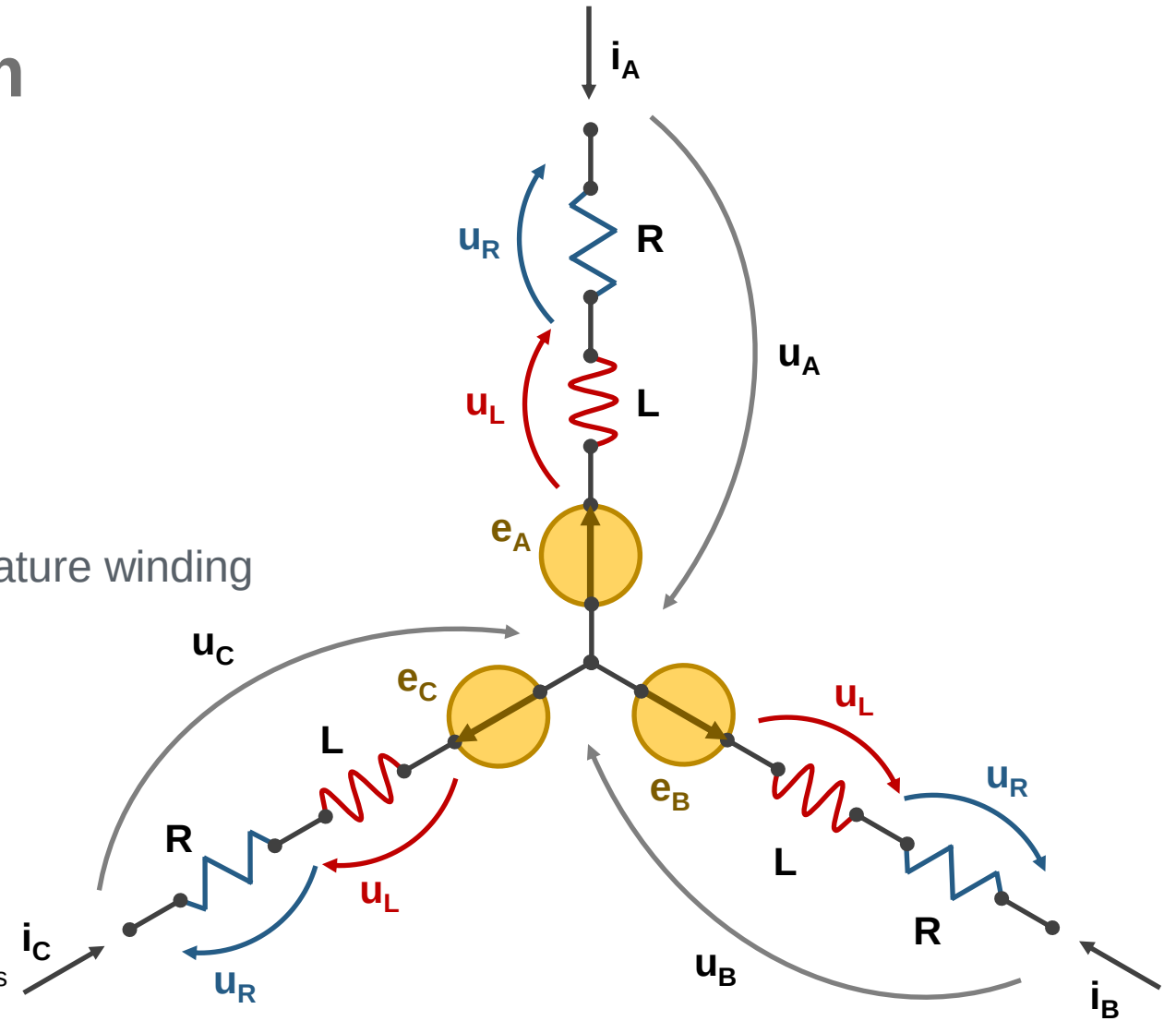
$$e_B = K_w \cdot f(\theta_e - 2\pi/3) \cdot \omega$$

$$e_C = K_w \cdot f(\theta_e + 2\pi/3) \cdot \omega$$

K_w – back-EMF phase constant

θ_e – electrical rotor angle equal with mechanical angle multiplied by number of poles

ω – rotor speed



L – armature self inductance

R – armature resistance

e – back-EMF

i – phase current

u – terminal phase voltage



BLDC Motor - Mechanical System

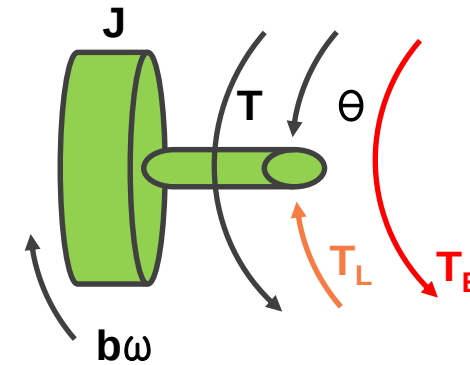
- Total Toque available at rotor shaft

$$T = T_E - T_L = J \cdot d\omega/dt + b \cdot \omega$$

- Electromagnetic Torque

$$T_E = (e_A \cdot i_A + e_B \cdot i_B + e_C \cdot i_C) \cdot \omega^{-1}$$

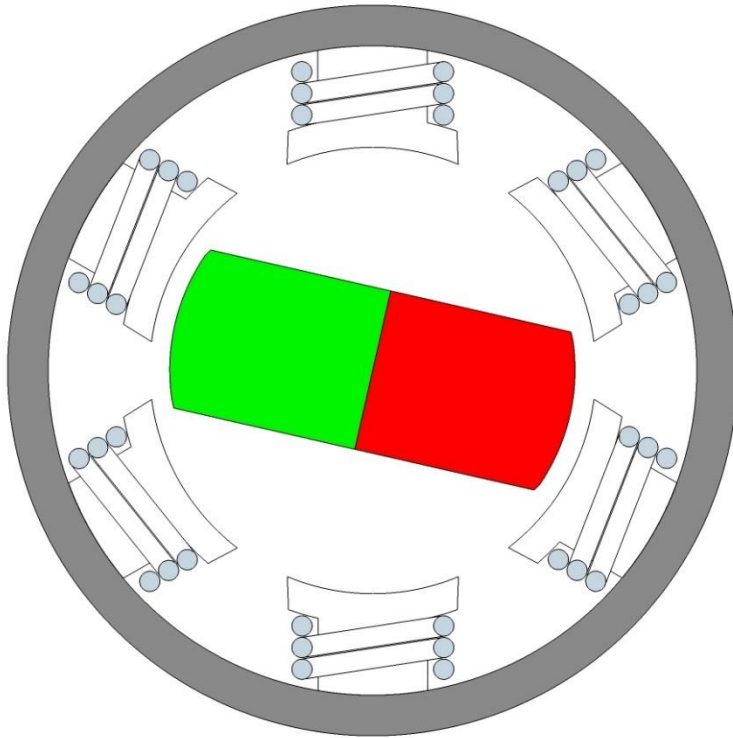
- T_L – is the load torque



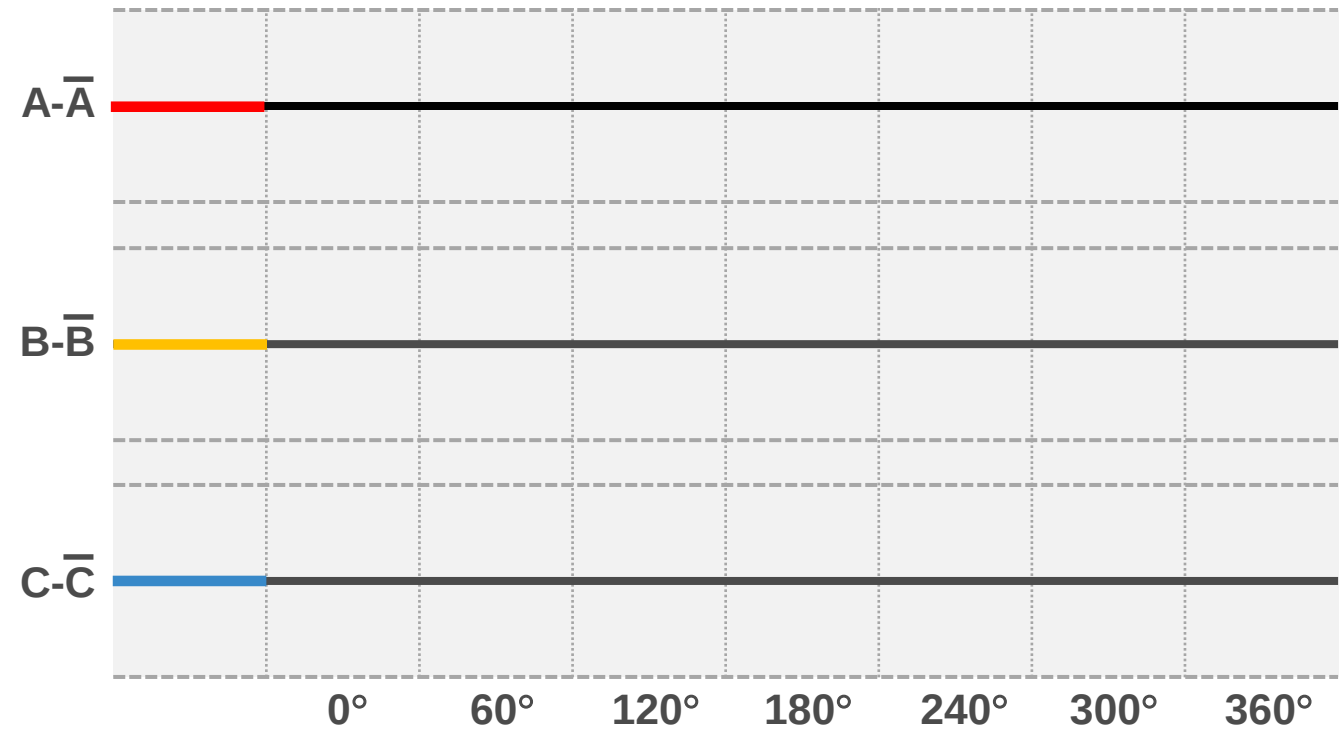
J – rotor inertia
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How it works

- Simplest commutation model is to energize one phase at a time to rotate the rotor CW or CCW

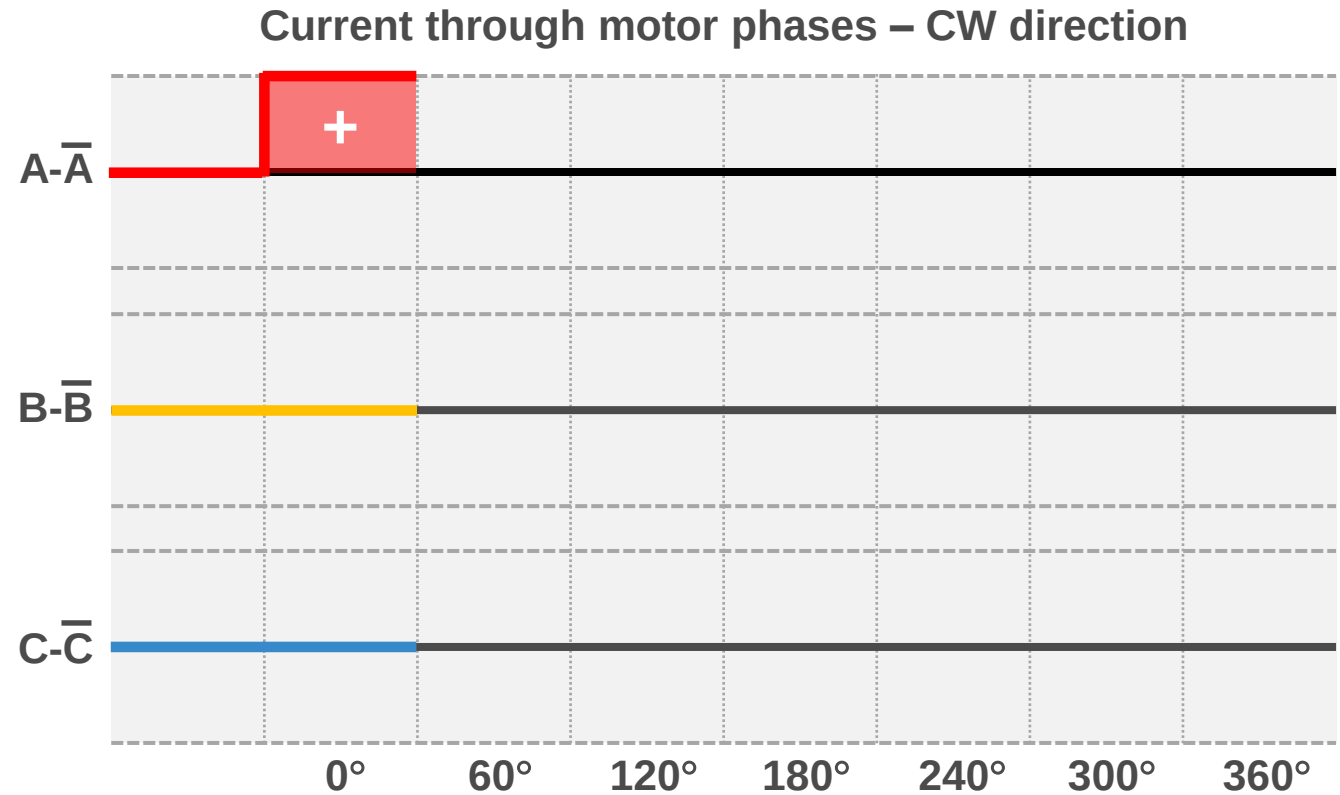
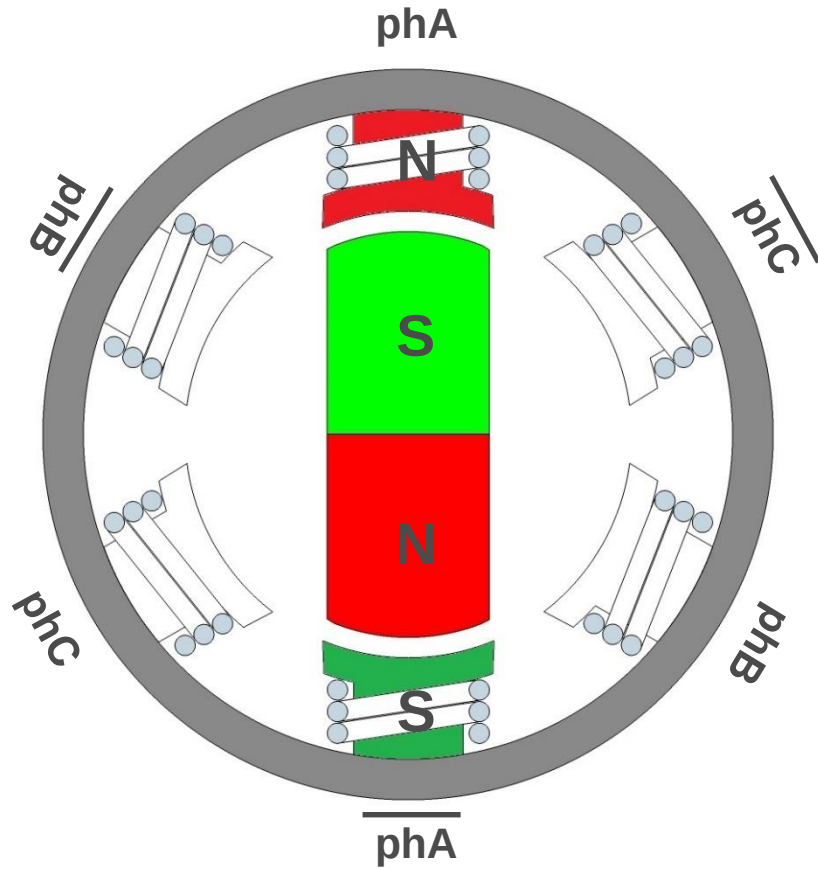


Current through motor phases – CW direction



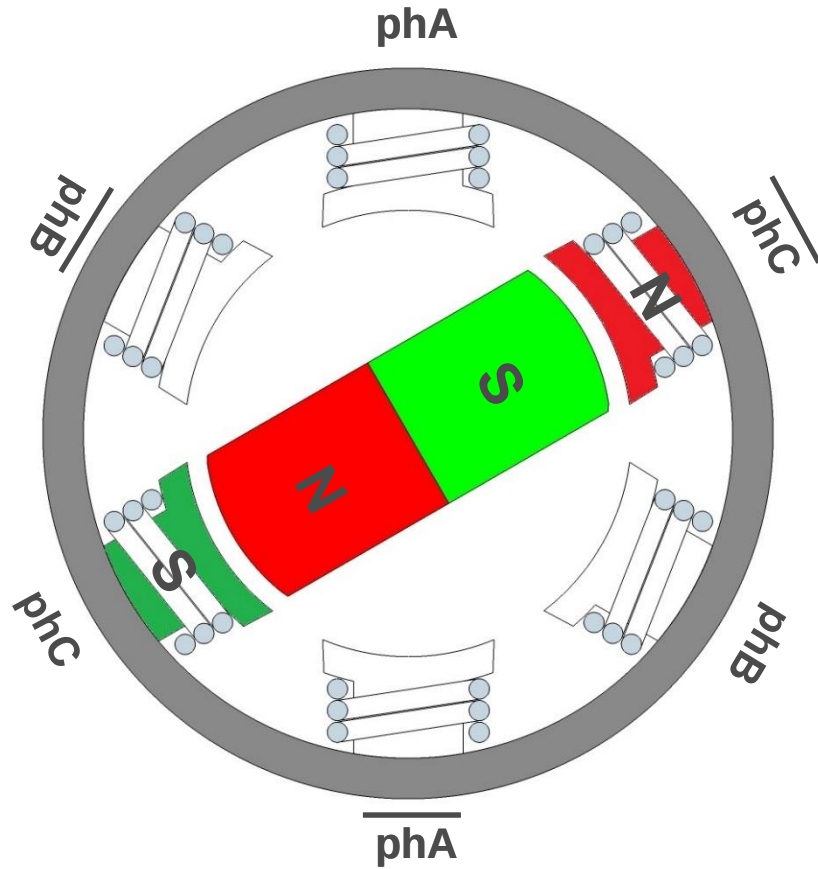
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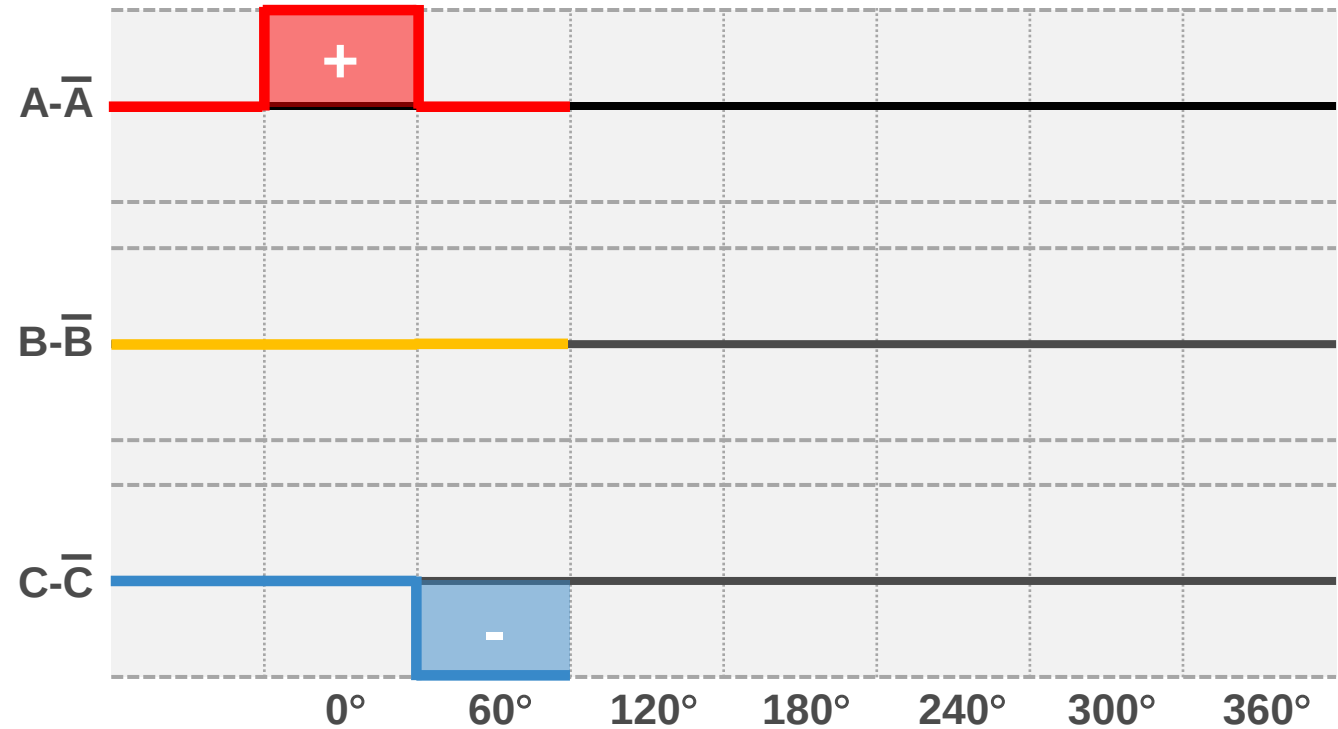


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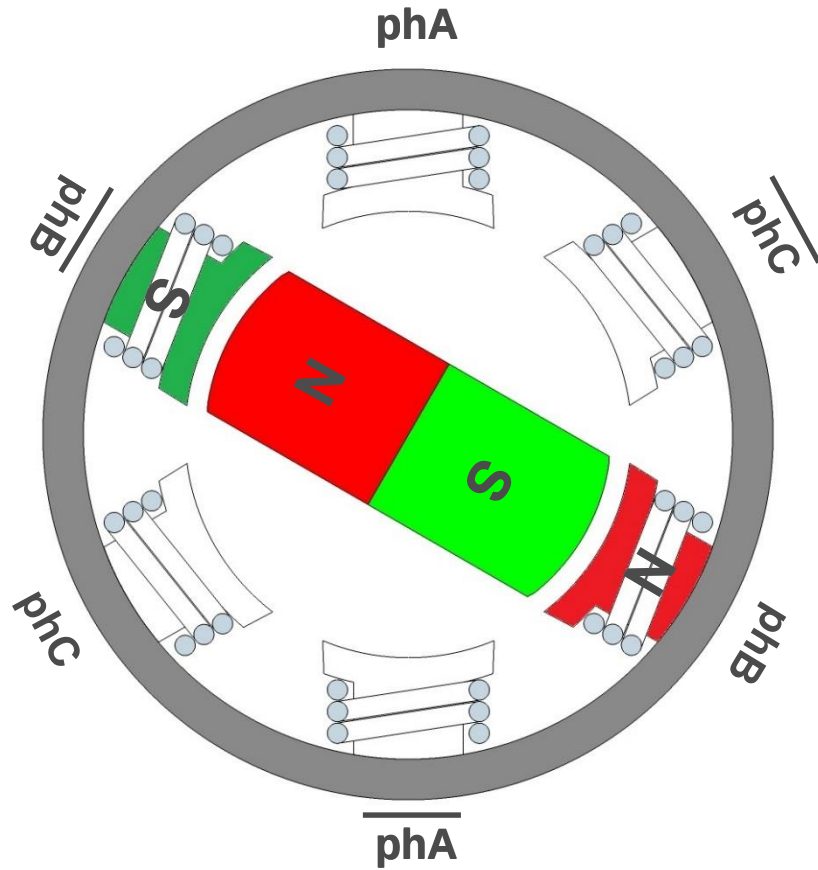


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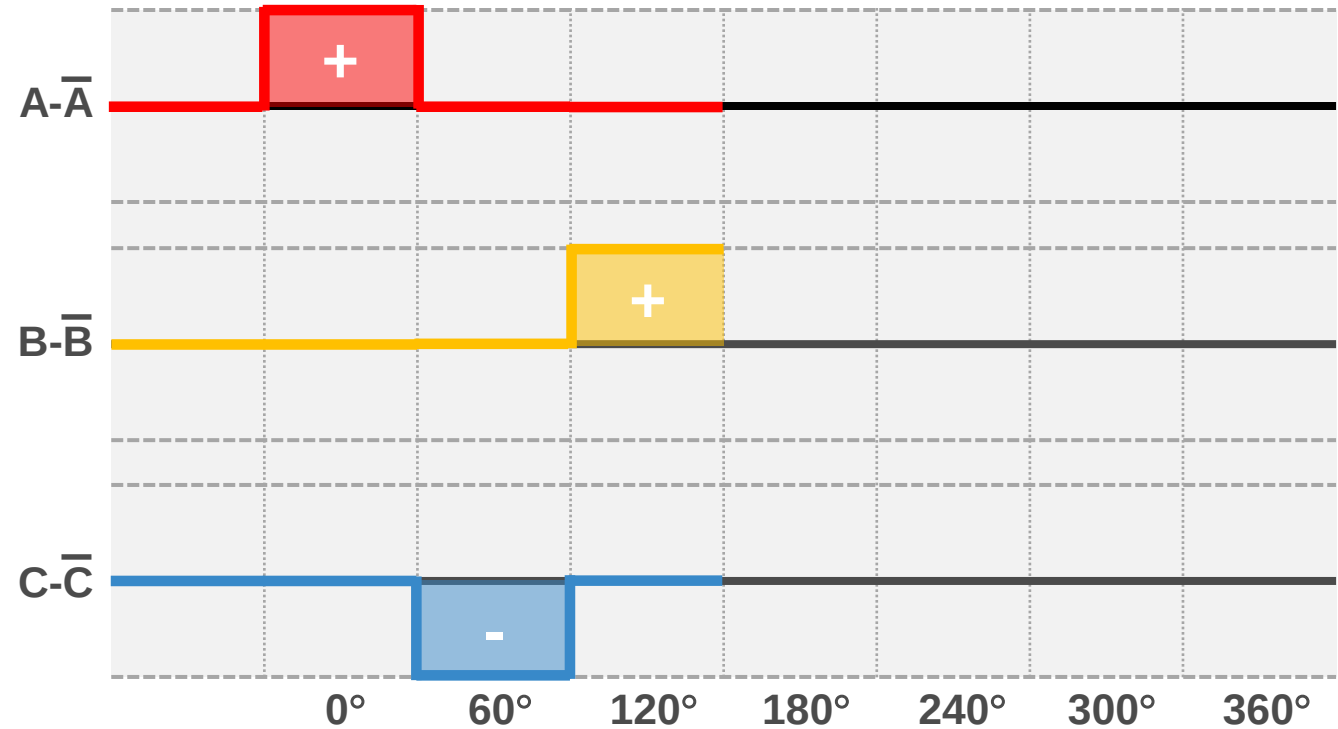


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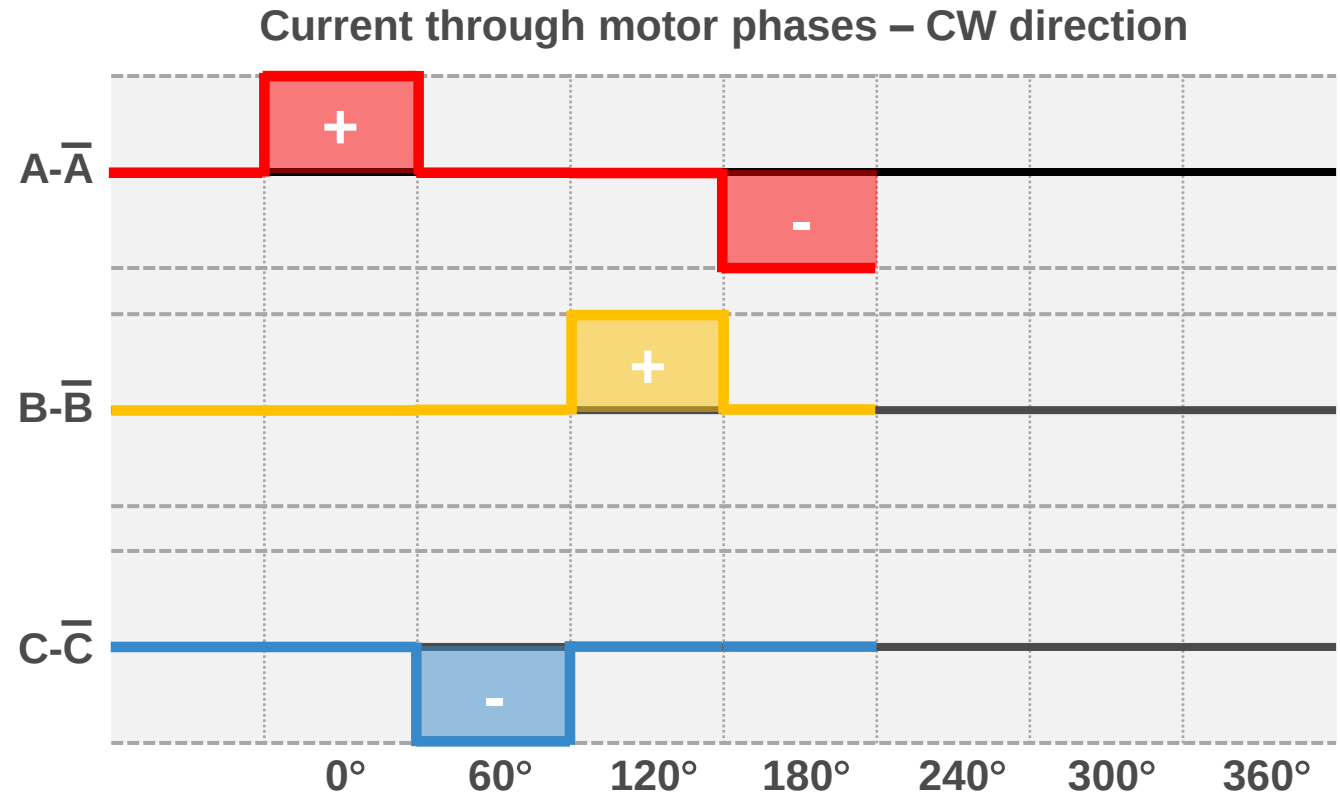
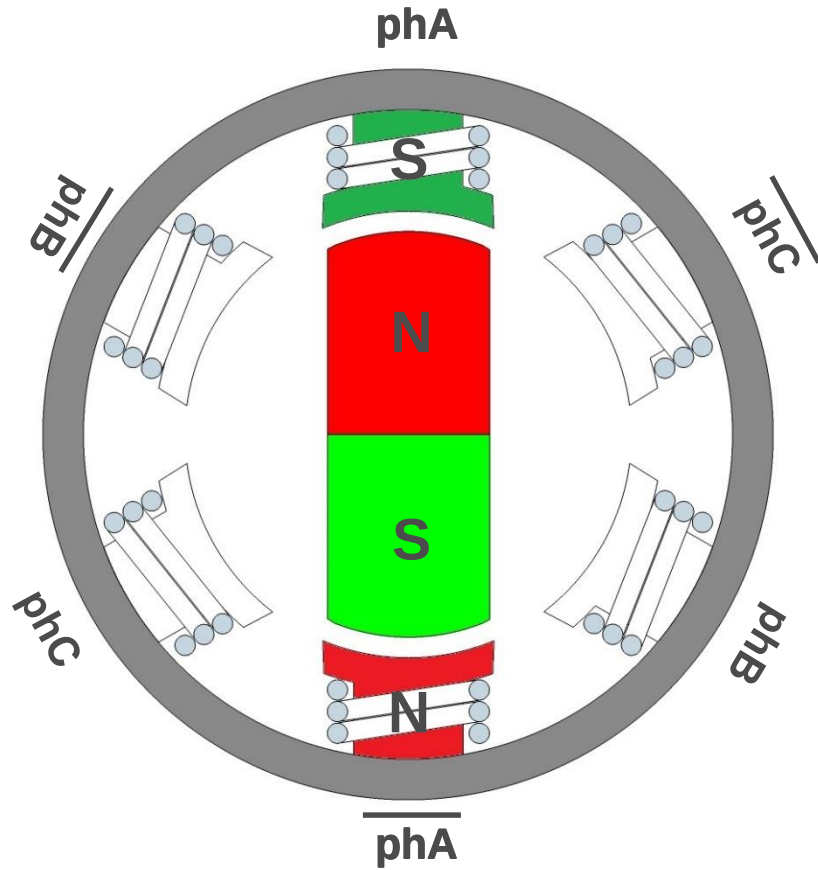


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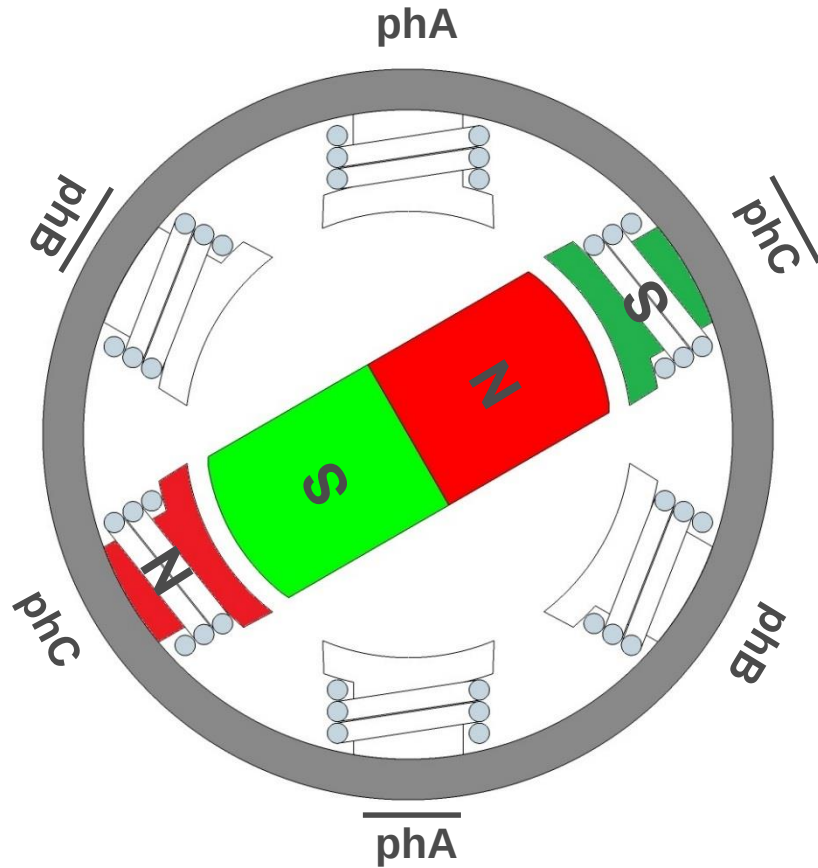
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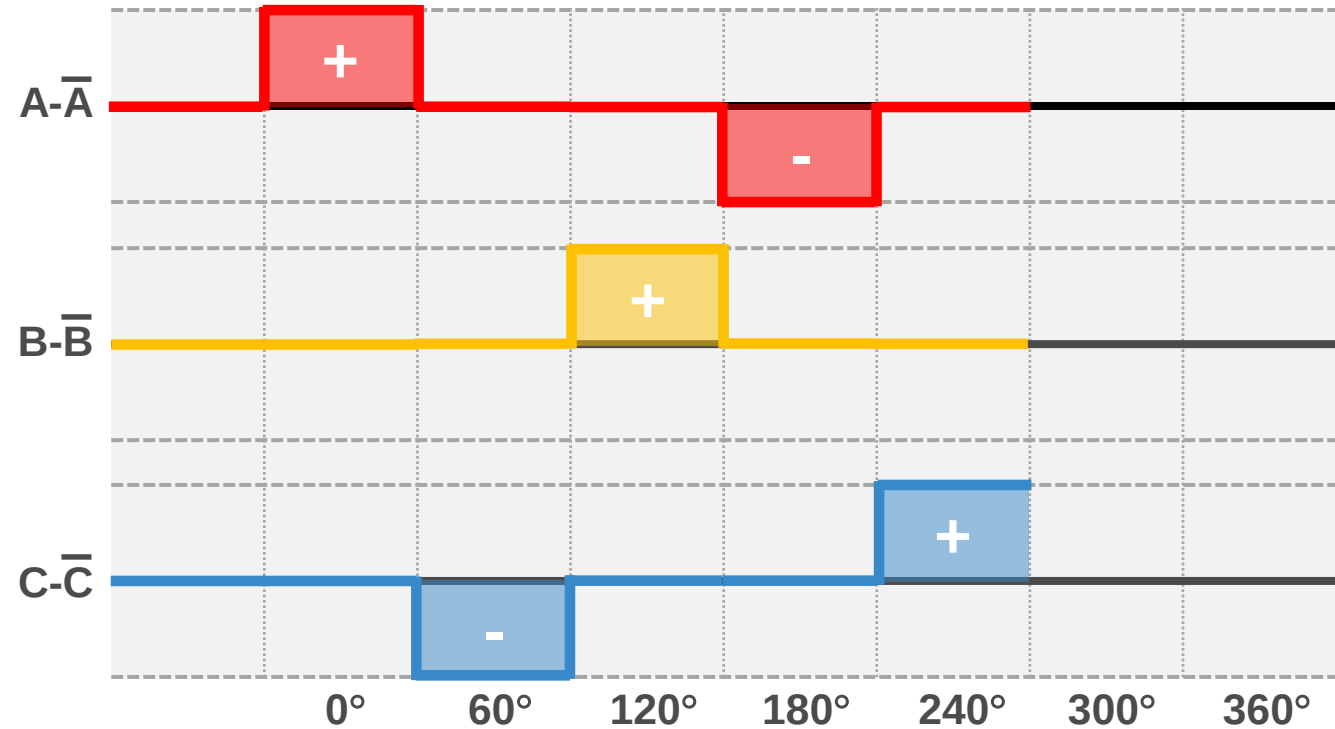


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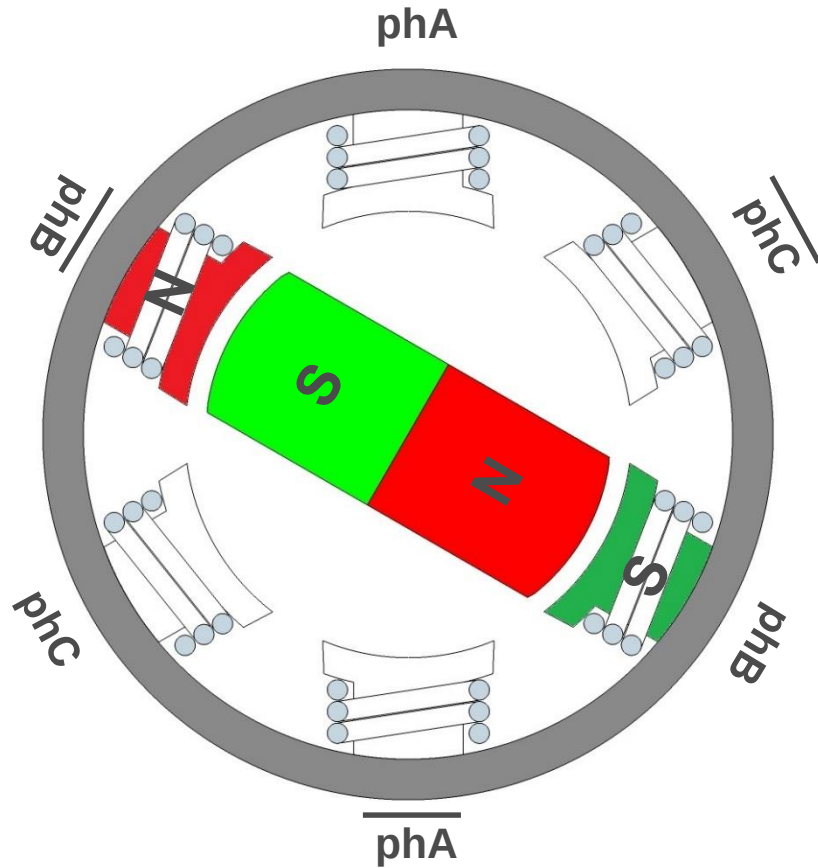


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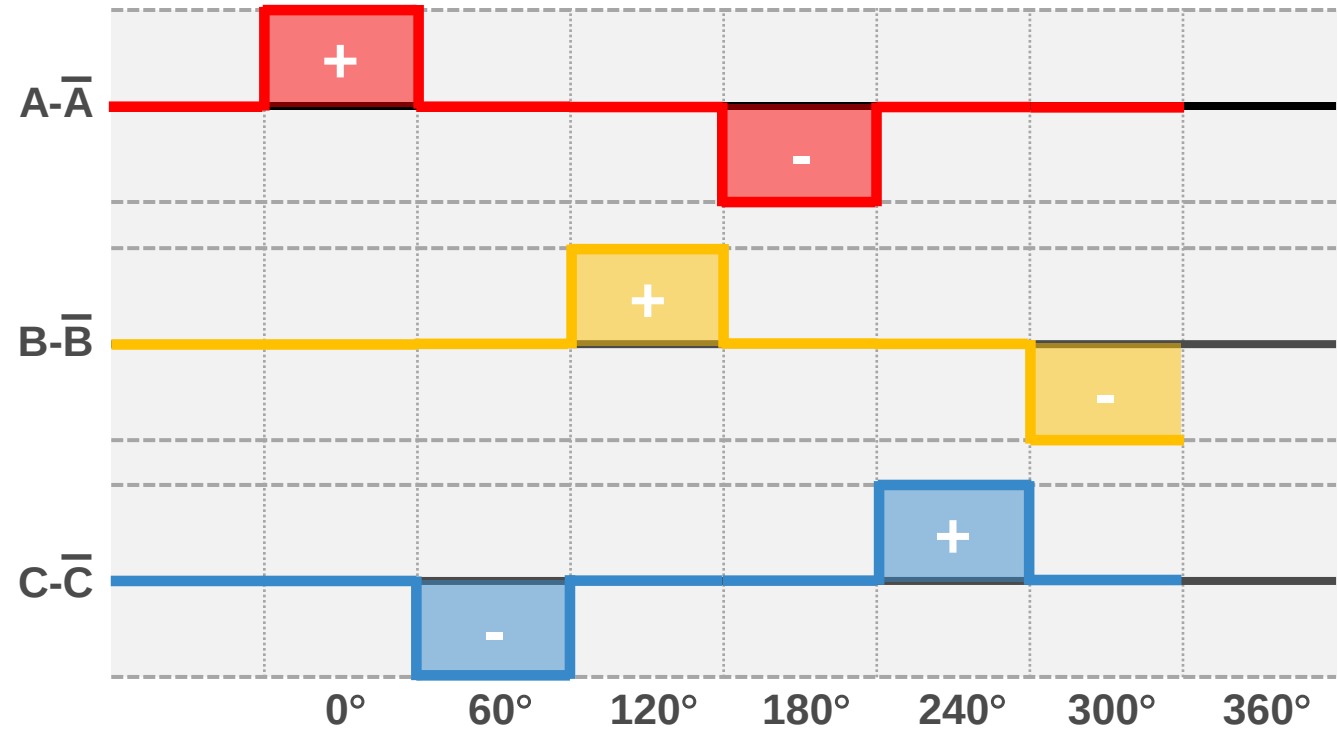


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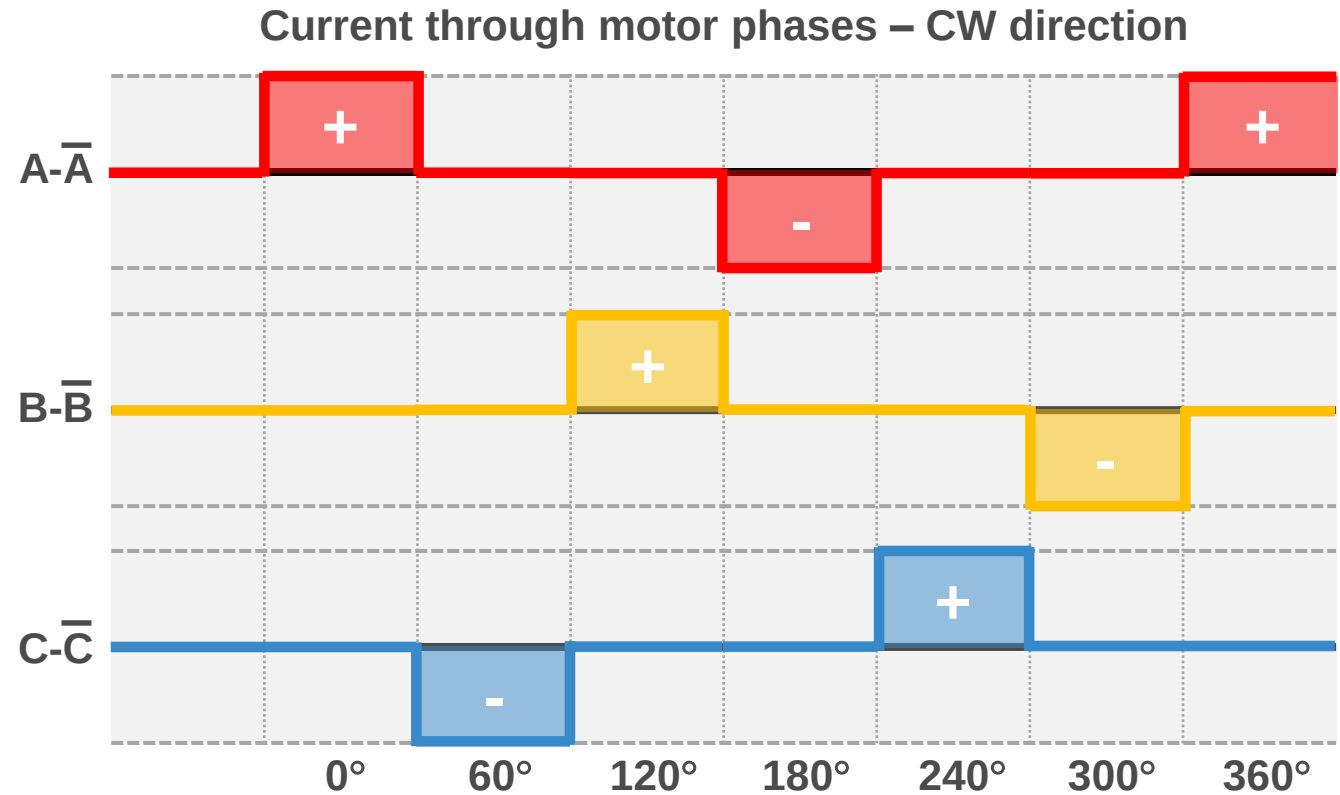
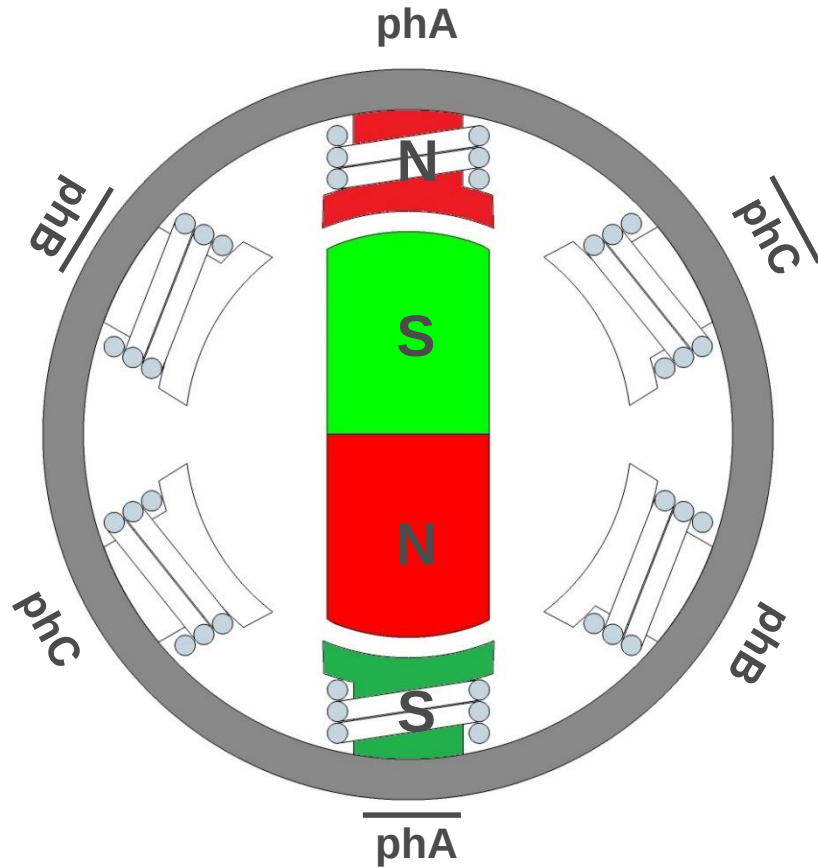


Current through motor phases – CW direction



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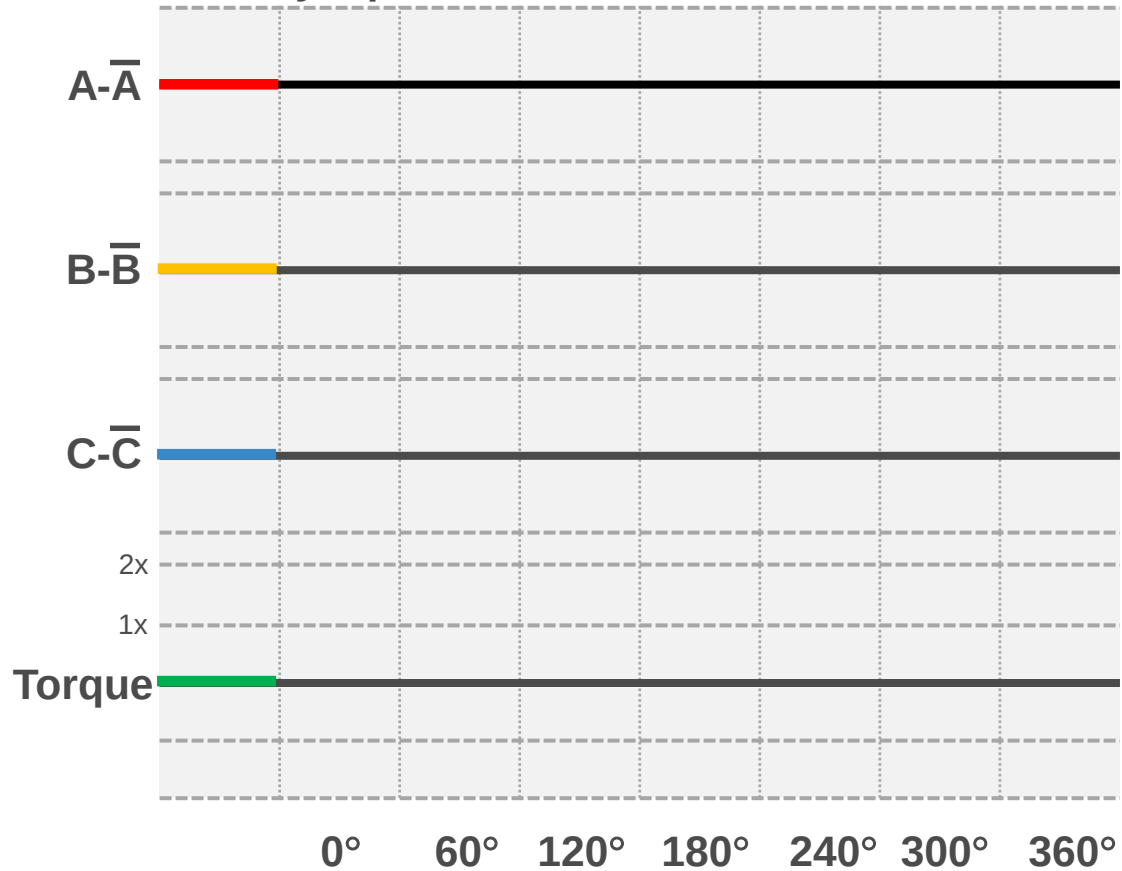
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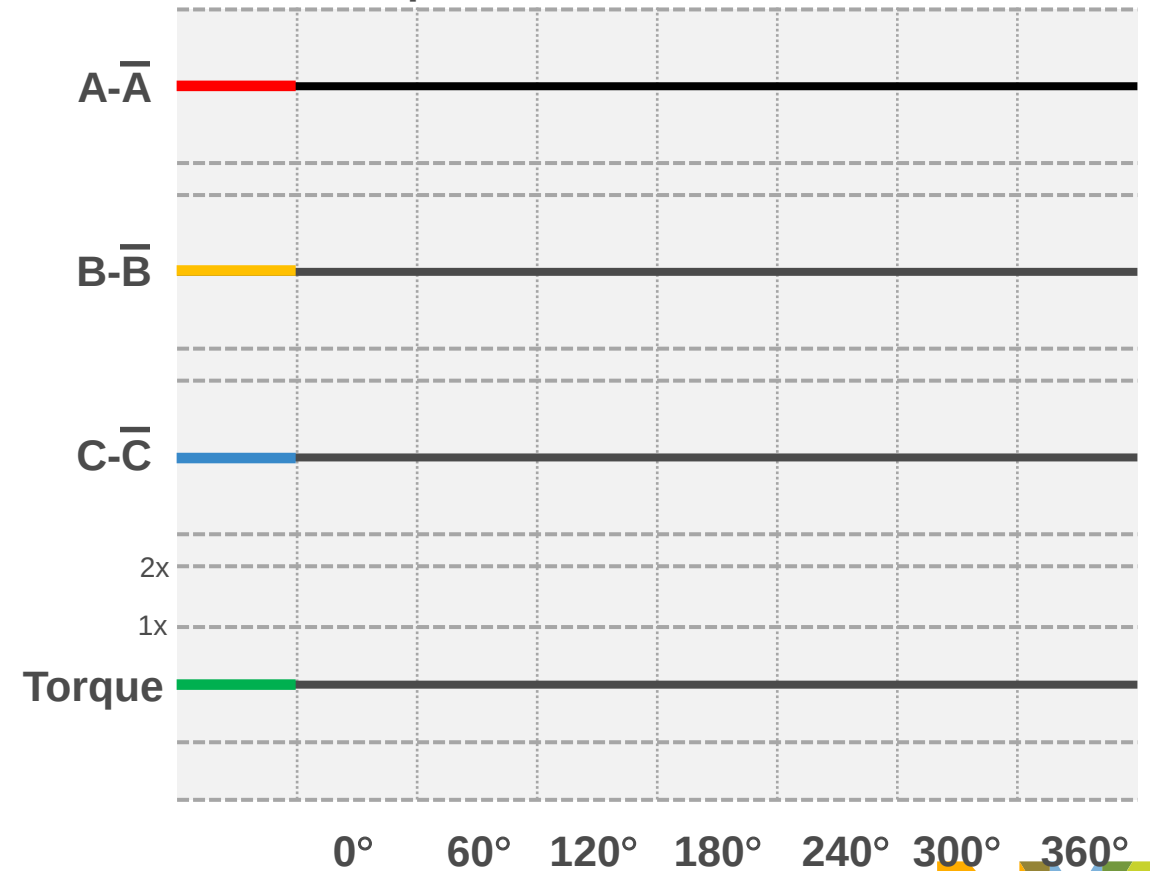
Practical Implementation

- Simplest commutation model is NOT efficient – only one phase out of 3 produces torque
- There is a better way – commute 2 out of 3 phase at each time

1 by 1 phase commutation CW use case



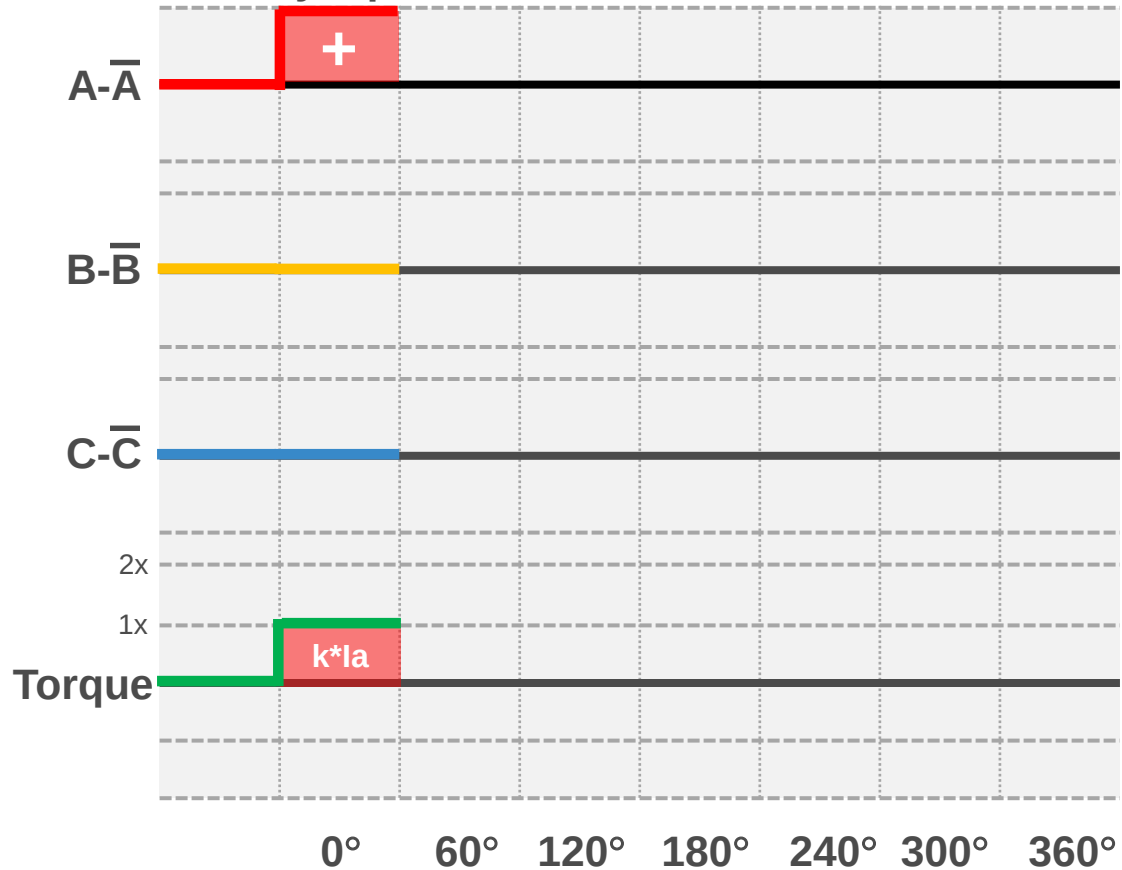
2 out of 3 phases commutation CW use case



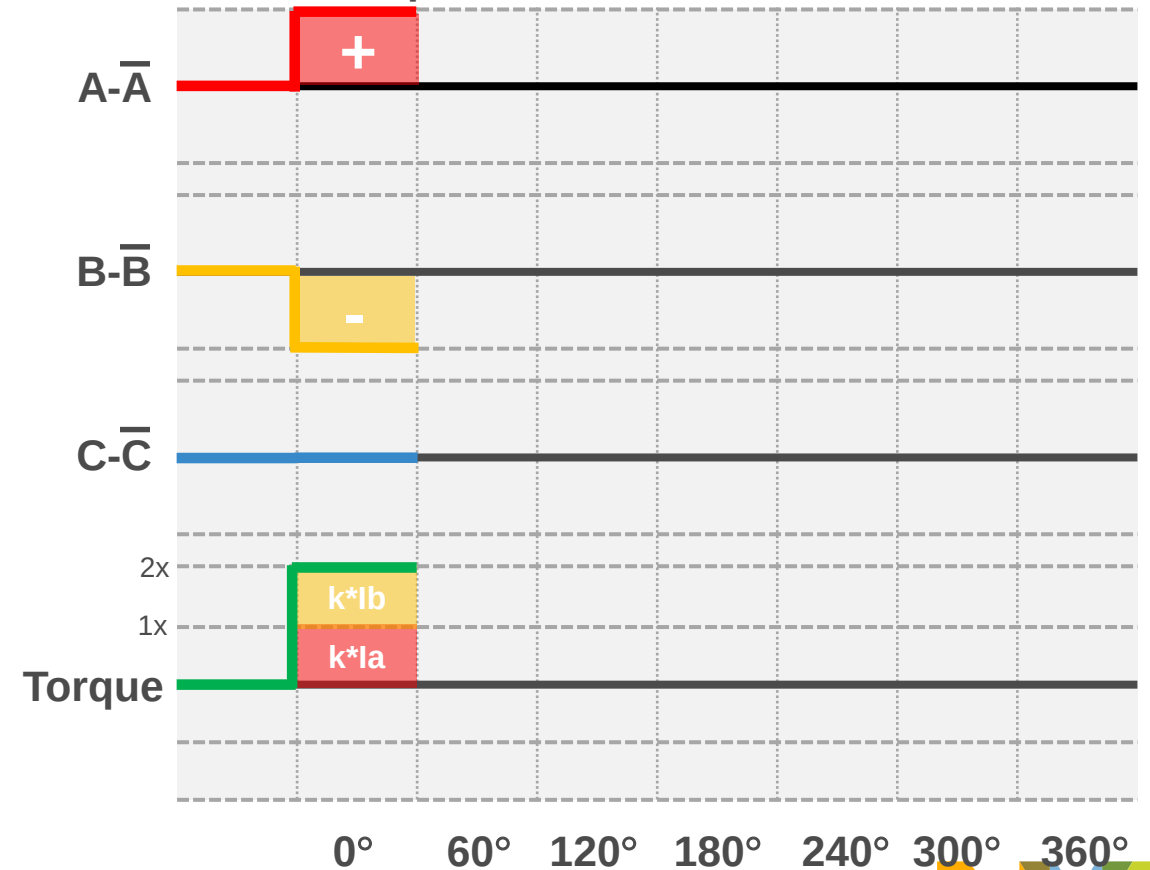
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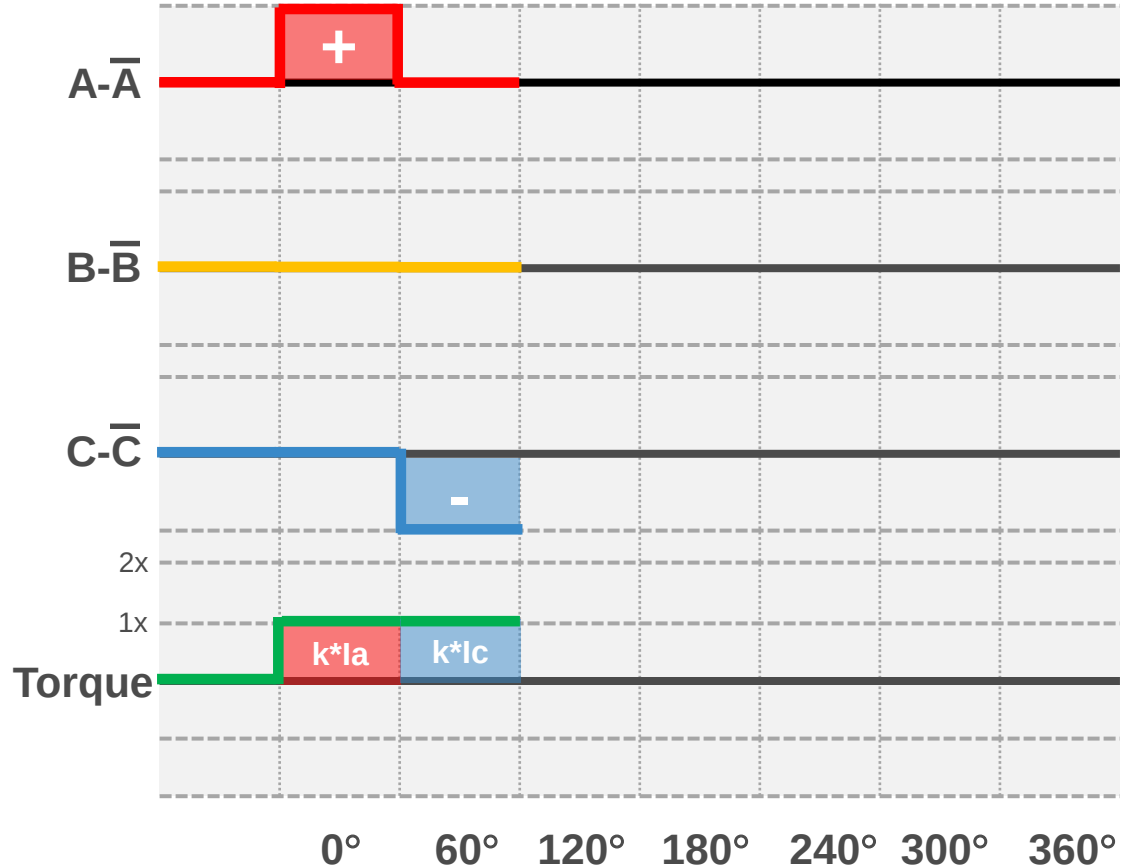
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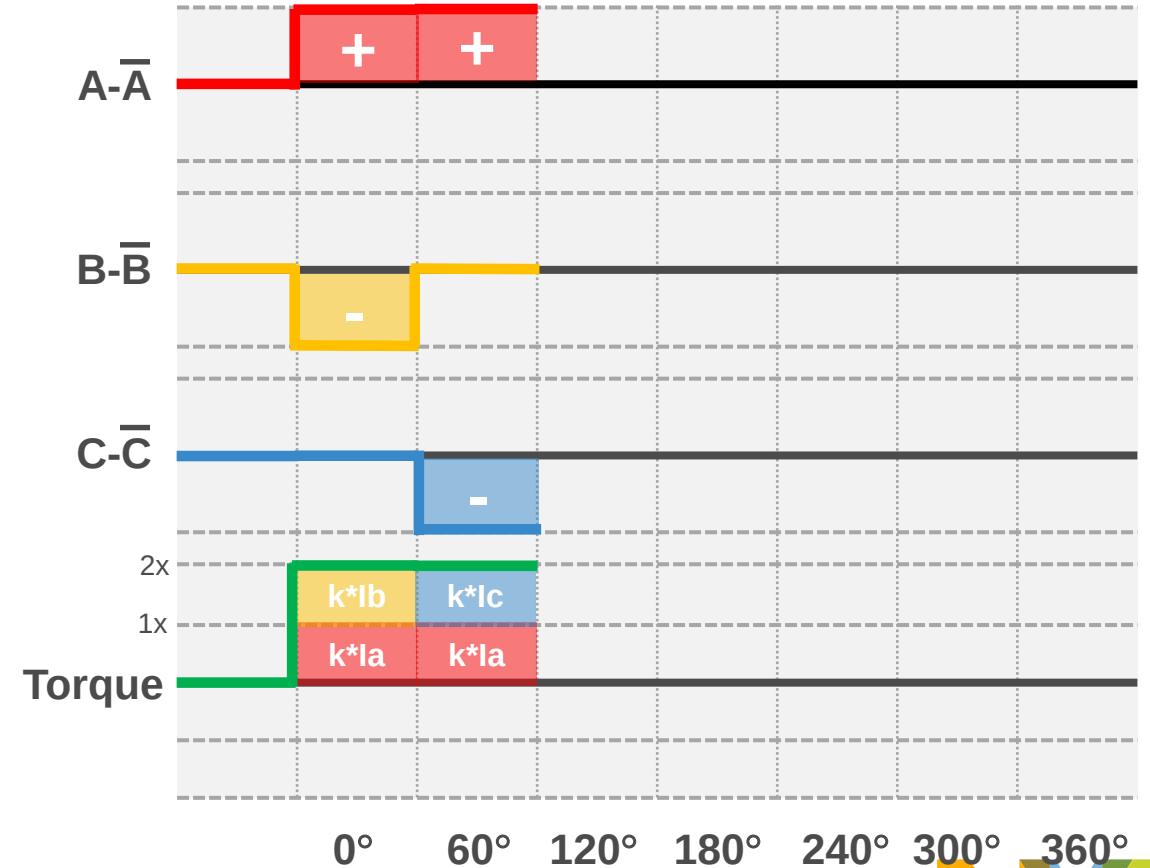
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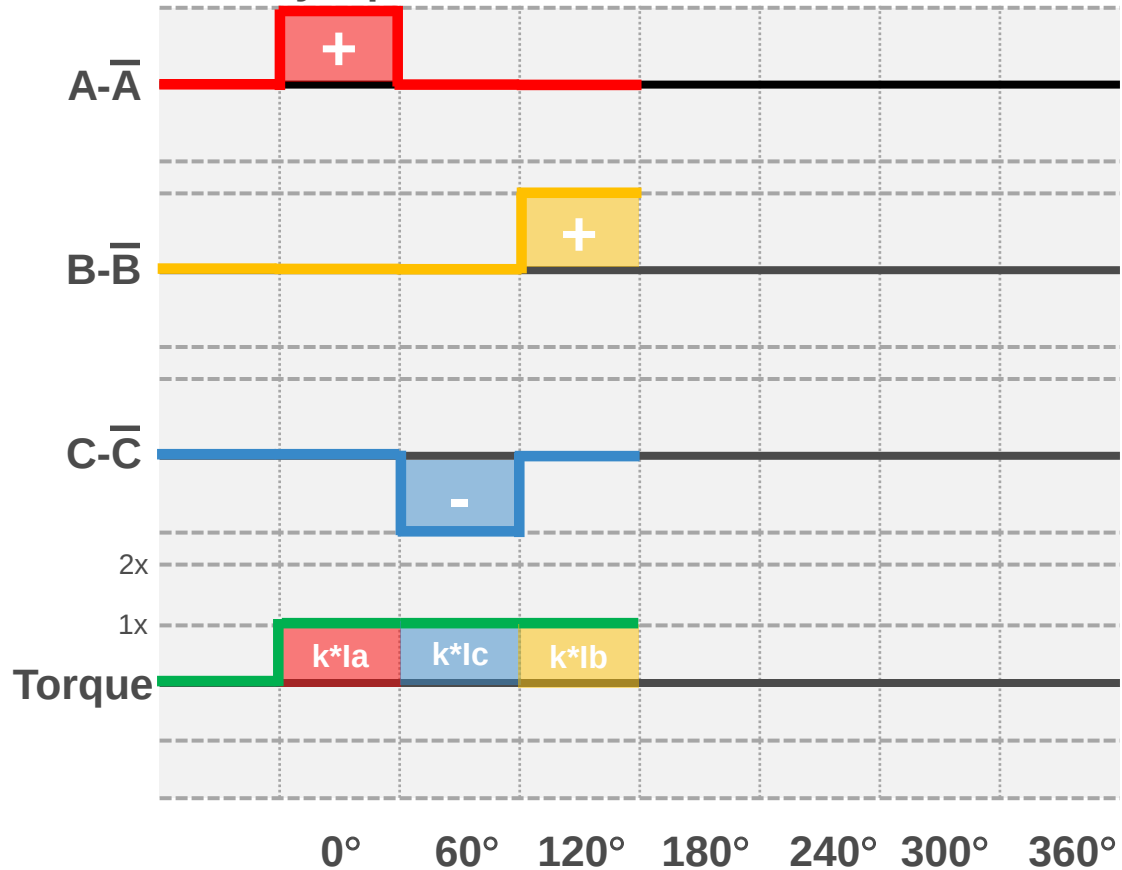
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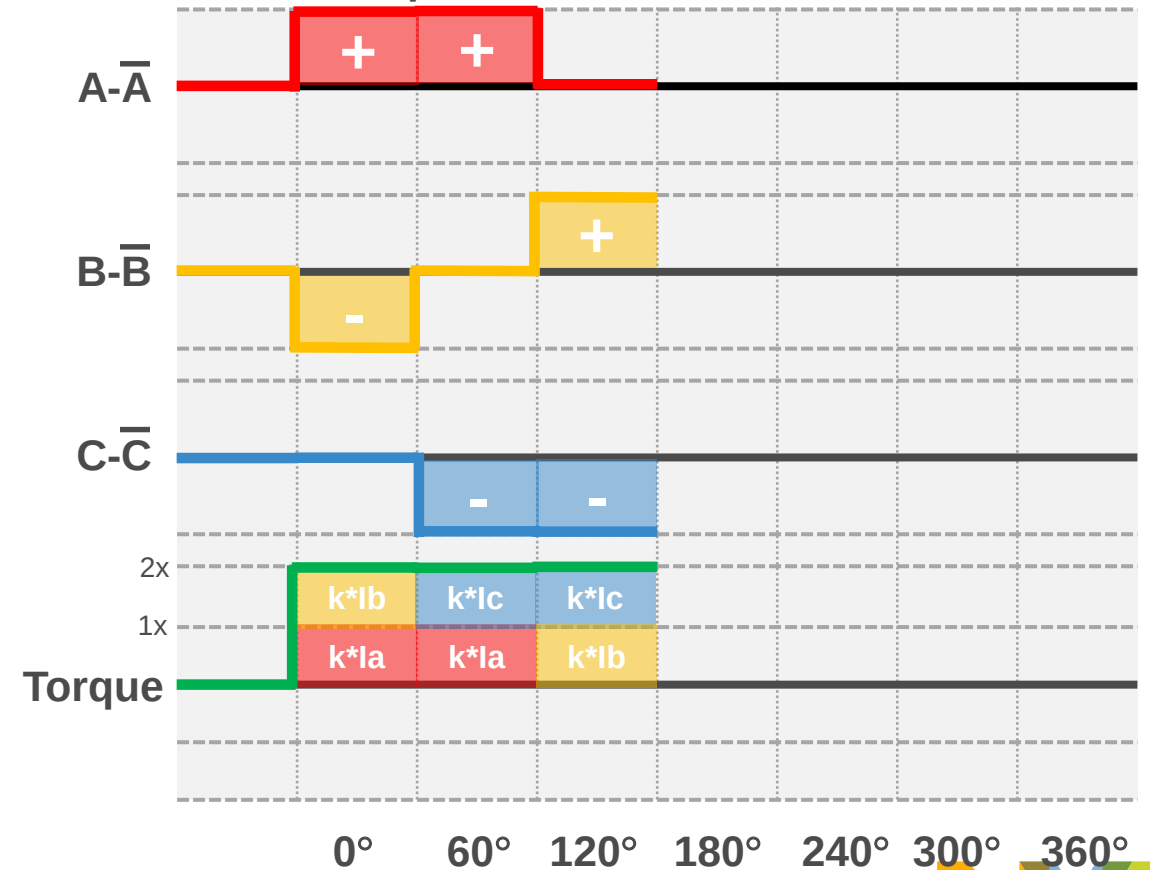
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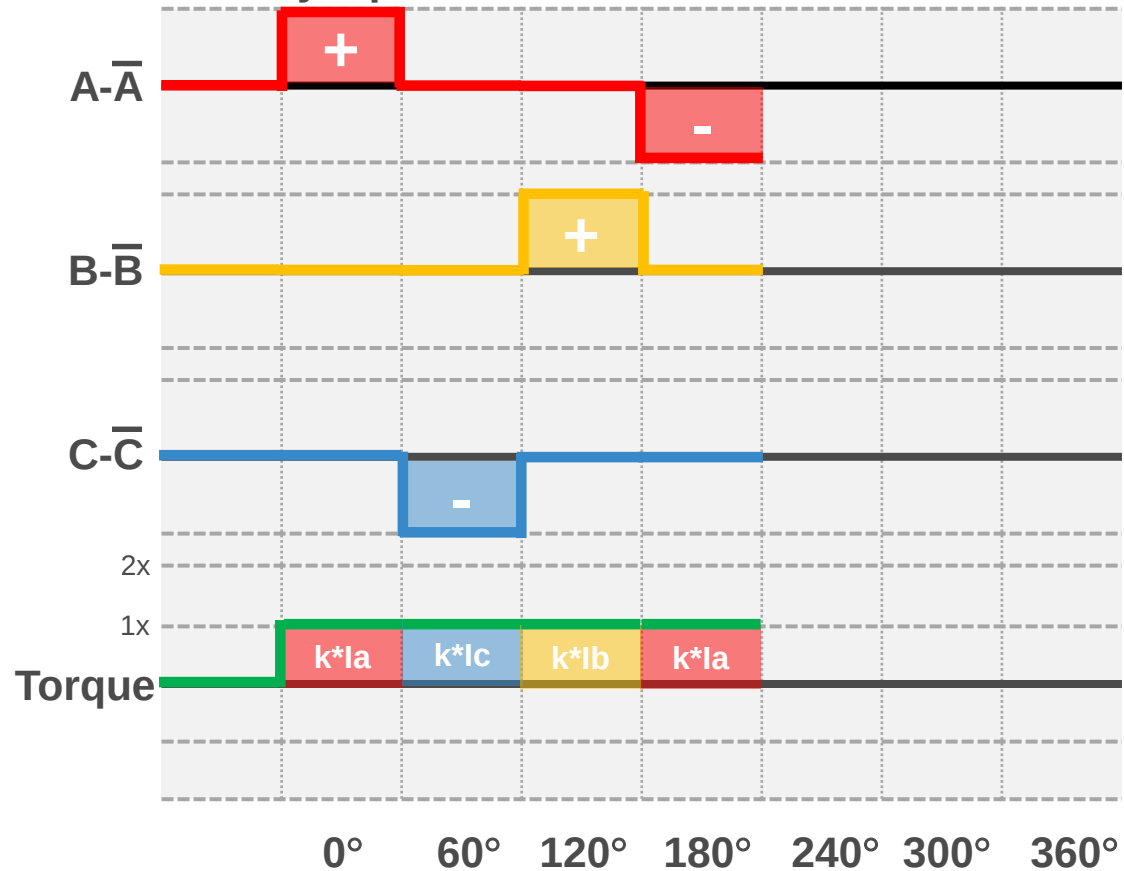
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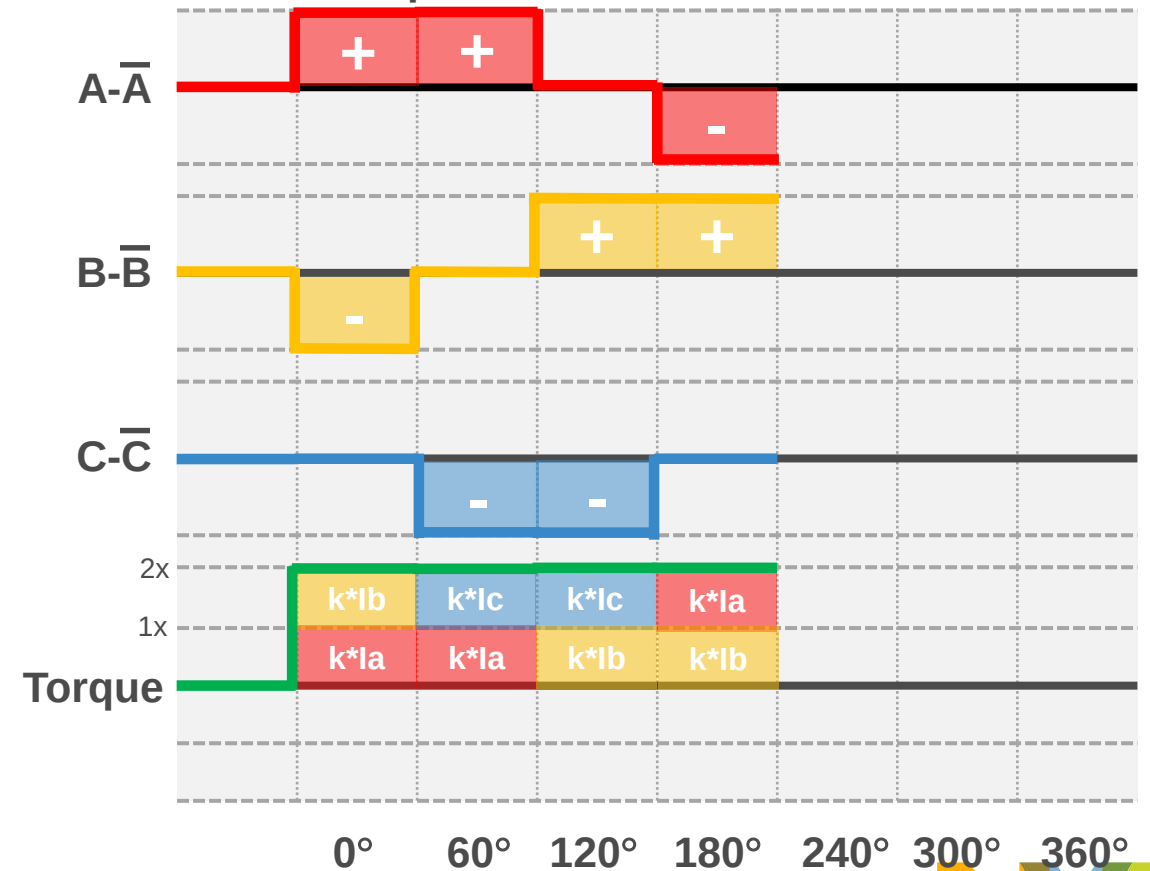
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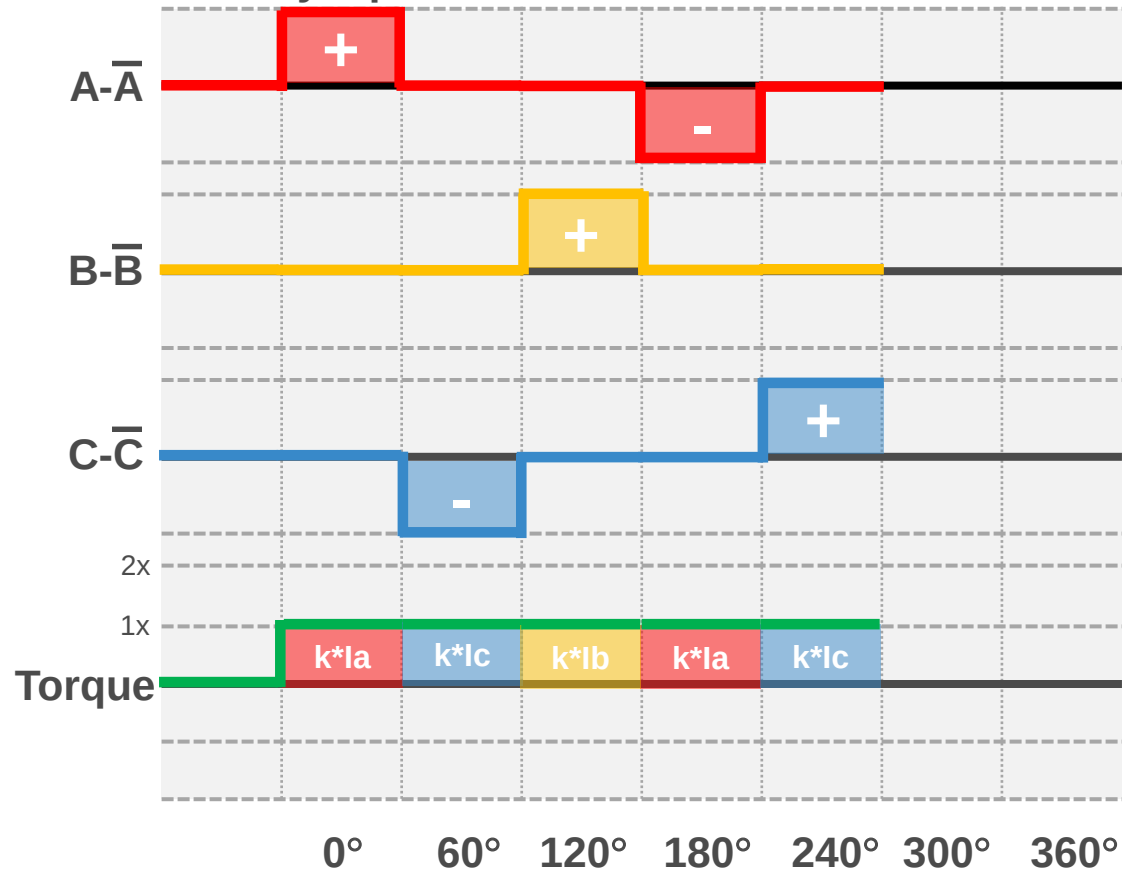
2 out of 3 phases commutation CW use case



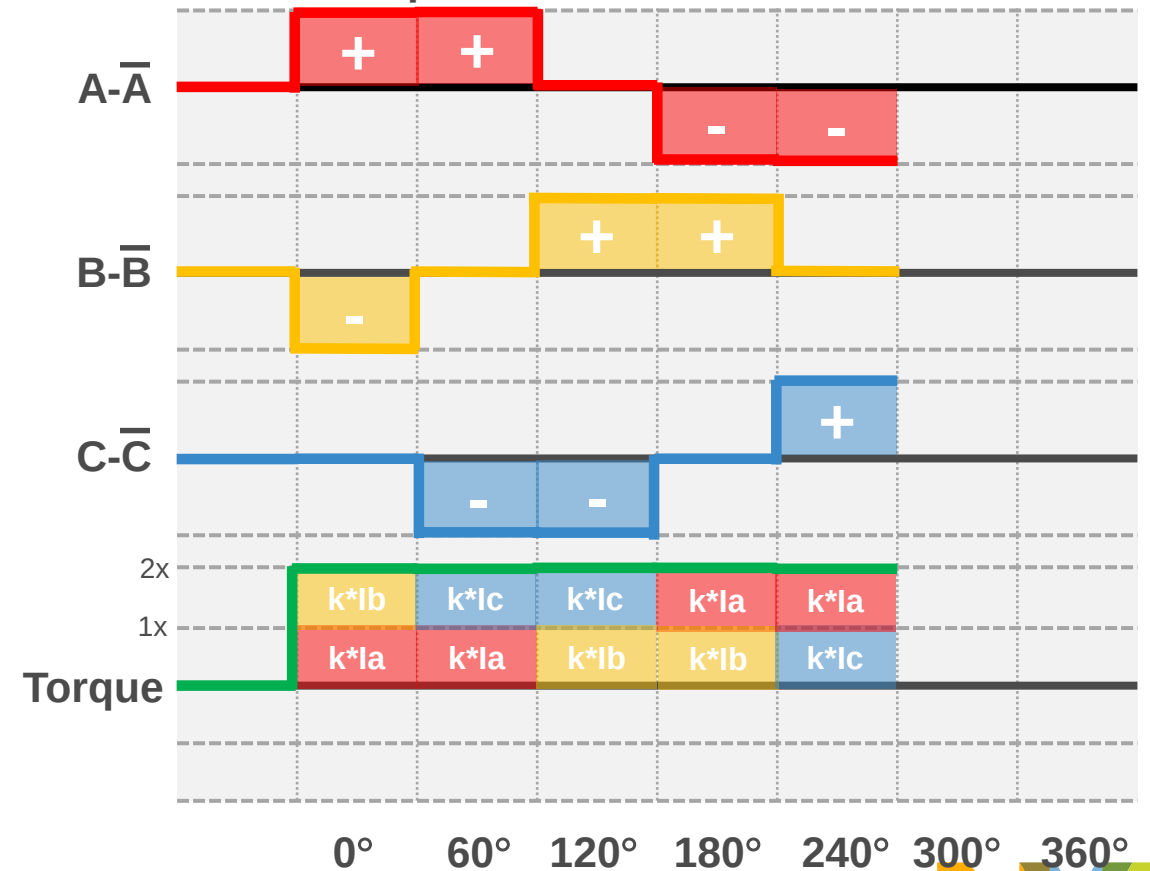
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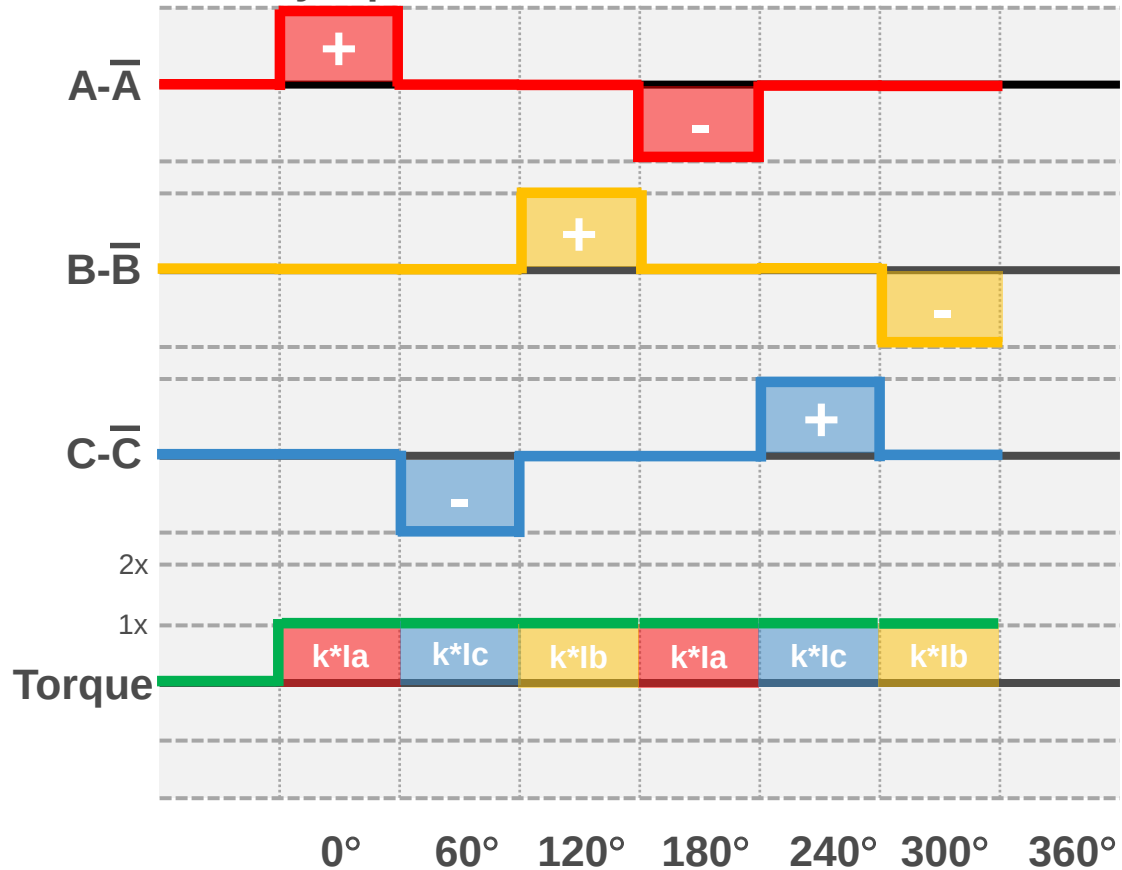
2 out of 3 phases commutation CW use case



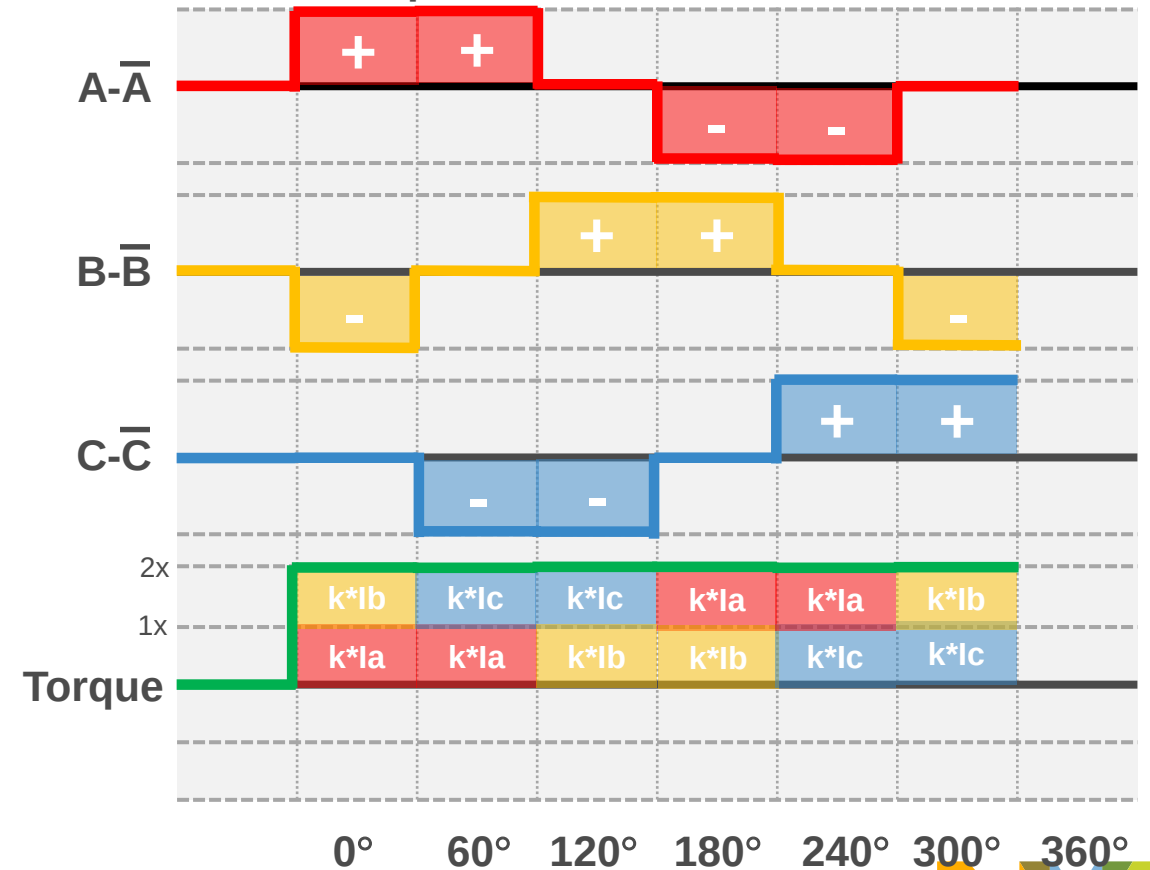
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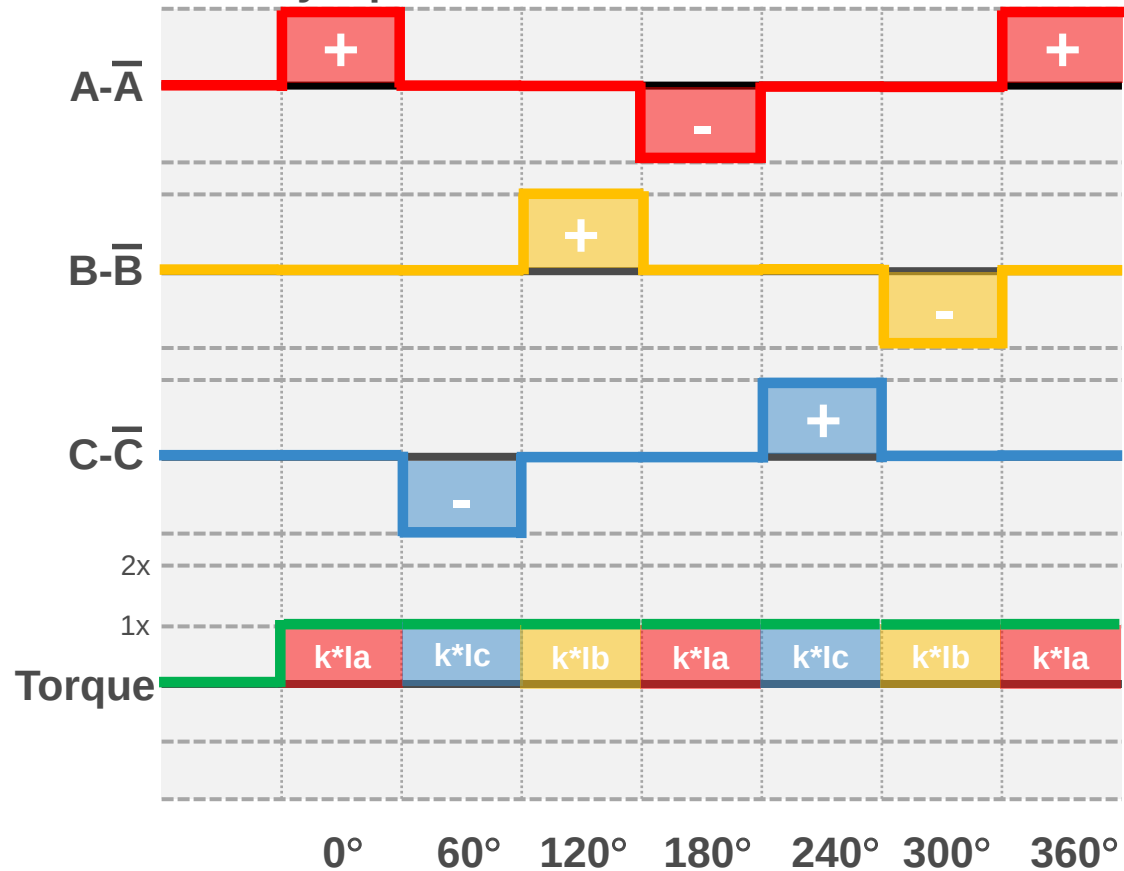
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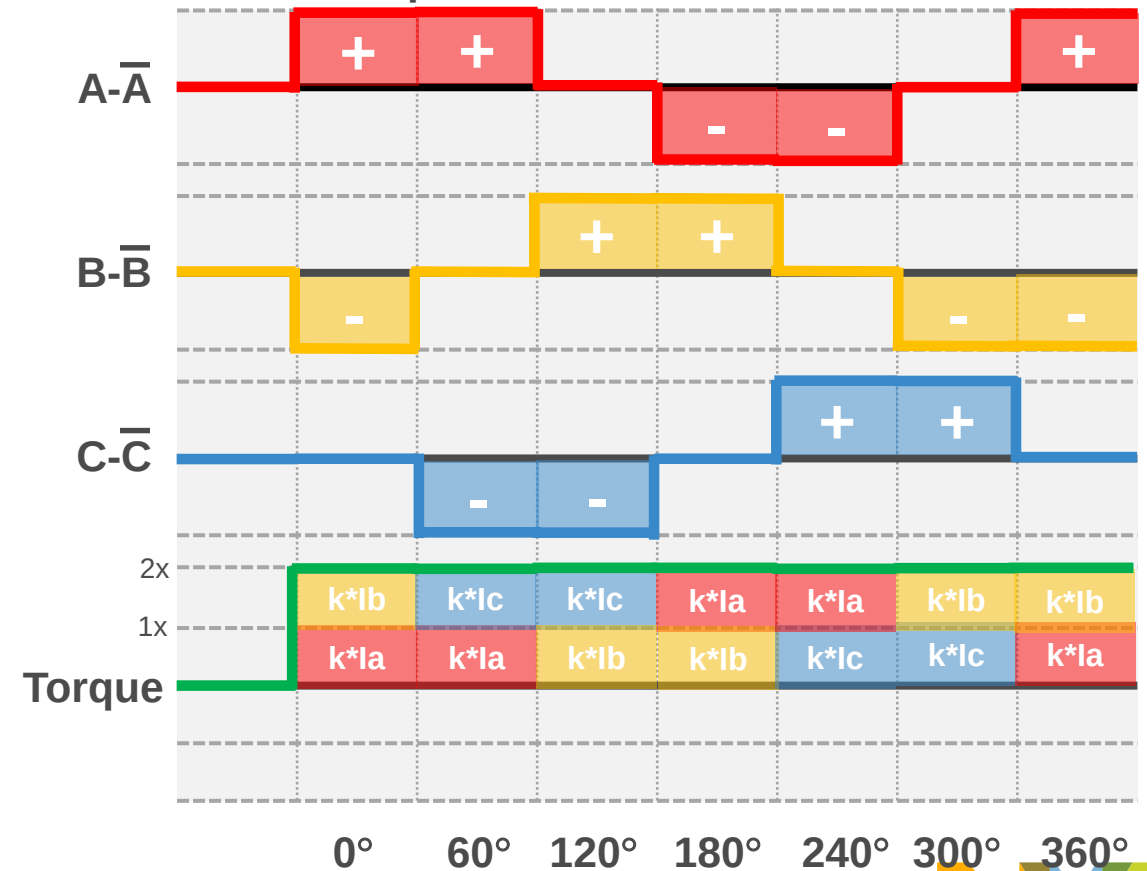
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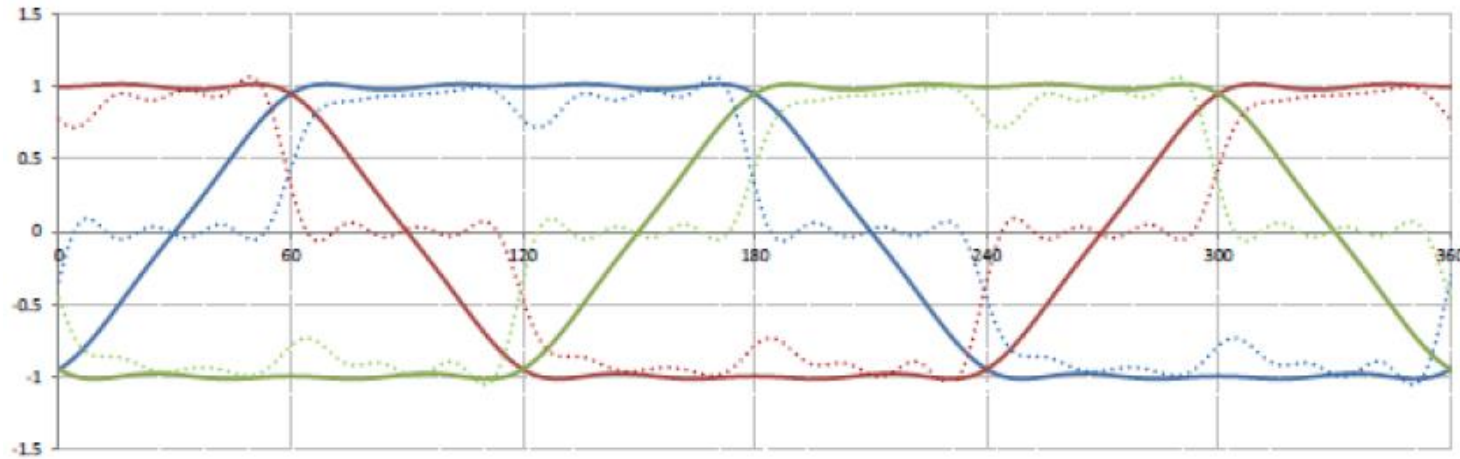


2 out of 3 phases commutation CW use case

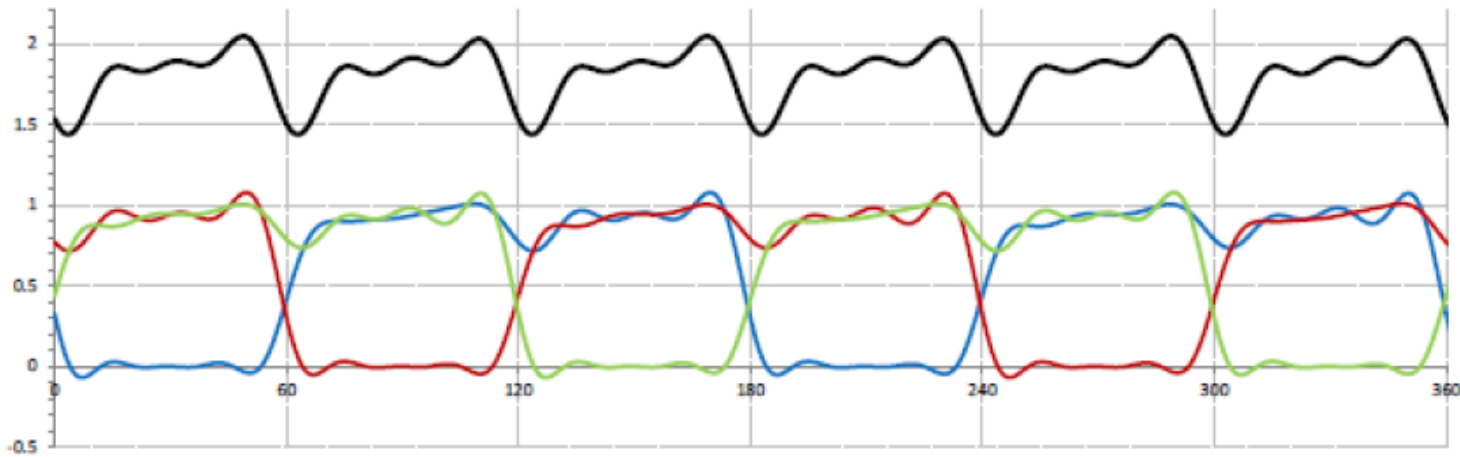


BLDC Motor – Torque Ripple

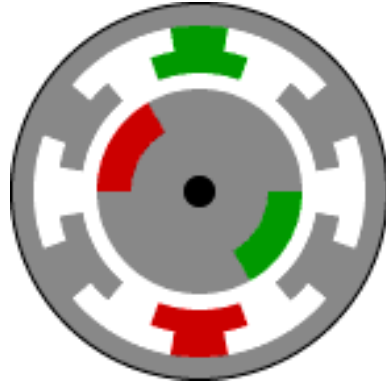
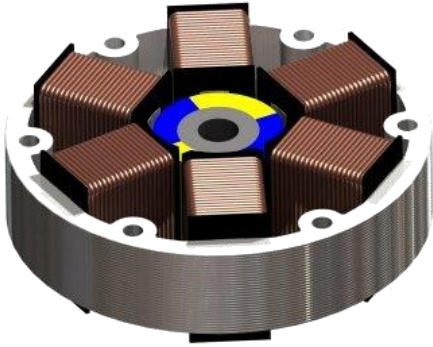
- Normalized back-EMF vs. Phase Currents (dotted-lines)



- Normalized Resulting torque (black) vs. Torque for each motor phases



Motor – BLDC vs. PMSM



BLDC motor

3-phase machine with PM on the rotor

Rotor position sensing required for rotor flux position

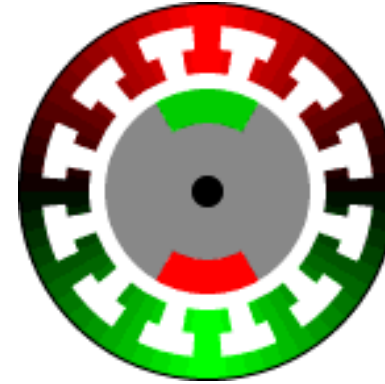
High torque per frame size

Synchronous operation

Trapezoidal Back-EMF

Six-step commutation control

High torque ripple



PMSM motor

3-phase machine with PM on the rotor

Rotor position sensing required for rotor flux position

High torque per frame size

Synchronous operation

Sinusoidal Back-EMF

Field-oriented control

Low torque ripple



MODEL BASED DESIGN TOOLBOX ENABLING FAST PROTOTYPING AND DESIGN

...you are now ready to design the next big thing. Good luck!

Need help? Visit us at: <https://community.nxp.com/community/mbdt> to share ideas

